

Consumer Flexibility, Data Quality and Targeted Pricing*

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Abstract

We consider a two-dimensional Hotelling model where consumers differ both in their brand preferences and transport cost parameters (flexibility). Firms hold perfect data on consumer brand preferences. We analyze their incentives to acquire data on consumer flexibility depending on its quality and on how strongly consumers differ along that dimension. We show that when consumers are weakly differentiated in flexibility, firms acquire data of any quality. In that case data acquisition gives rise only to a rent-extraction effect, increases profits and harms consumers. When consumers are strongly differentiated in flexibility, data acquisition intensifies competition and both firms acquire flexibility data only if its quality is good enough. While firms lose from data acquisition, consumers are better off. Our paper contributes to the literature on competitive price discrimination with demand-side asymmetry. We consider a model with best-response asymmetry (Corts, 1998) and show that the sign of the profit effect of price discrimination based on consumer flexibility depends on consumer heterogeneity along that dimension.

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1 Introduction

Firms in many industries store, collect and analyze customer data, which allows them to conclude on consumer preferences and price discriminate among customers respectively.¹ Both theoretical and empirical research often distinguish between consumer brand preferences and the strength thereof (often referred to as “transport cost parameters”) giving rise to two types of customer data.² While much is known on the profit effects of price discrimination based on consumer brand preferences, there is relatively little research on the profit effects of price discrimination along consumer transport cost parameters (flexibility). The latter is the focus of our paper together with the related question of firms’ incentives to acquire data on consumer flexibility.

We consider a Hotelling model with two-dimensional consumer heterogeneity: along consumer brand preferences and the strength thereof. We assume that firms hold perfect data on consumer brand preferences and, hence, can practice perfect third-degree price discrimination along that dimension.³ Additionally, firms can acquire data on consumer flexibility of the exogenously given quality. We analyze firms’ incentives to acquire customer flexibility data depending on its quality and on how strongly consumers differ in flexibility. For instance, our model describes well spatial competition between retailers. Following a boom of location-based marketing enabled by the widespread use of smartphones with built-in GPS devices, retailers are able to target consumers based on their precise geographic position (see, for instance, Beard, 2011).^{4,5} However, location data is nowadays by far not the only type of customer information

¹Loyalty programs, which are very widespread in airline and retail industries, provide a very important source of customer data. The CEO of Safeway Inc., the second-largest supermarket chain in the U.S., Steve Burd, said that “*There’s going to come a point where our shelf pricing is pretty irrelevant because we can be so personalized in what we offer people*” (Ross, 2013).

²For instance, Allenby and Rossi (1999) use a continuous model of consumer heterogeneity to estimate the distribution of consumer brand preferences and price sensitivities based on a scanner panel dataset of ketchup purchases. Kim, Srinivasan and Wilcox (1999) estimate price sensitivity of each household in a similar scanner panel data. They conclude that customer data collected by retailers within loyalty programs (which mainly includes data on consumers’ shopping patterns) allows to predict better consumer price sensitivity than demographic data and, hence, provides a better guidance for retailers in targeting individual consumers with price promotions.

³This assumption allows us to concentrate on the effect of price discrimination along consumer flexibility.

⁴“Geo-conquesting” is becoming an extremely popular strategy in mobile marketing, where a firm targets prospective customers when they are close to the competitor’s location (see, for instance, Tode, 2013b). In that case while designing its targeted offers a retailer takes into account not only the distance of prospective consumers to its own location, but also to that of the rival, which calls for the analysis of firms’ targeting activities in a model of spatial *competition*.

⁵Several mobile phone applications (such as 8Coupons and Mobiqpons) rely on the GPS devices to transmit real-time information on the physical location of a consumer and allow retailers to custom-target rebates respectively (see Wortham, 2009).

used in location-based marketing. Huge customer databases collected by mobile network operators provide an additional opportunity for targeted pricing yielding data on consumer transport cost parameters. In the recent years mobile firms started selling this data to marketing firms, advertisers or other interested parties raising concerns of privacy advocates (see, for instance, Tene and Polonetsky, 2012, and Tode, 2013a).^{6,7}

Our article contributes to the literature on competitive price discrimination with demand-side asymmetry. In that case consumers can be classified into different groups depending on their preferences for a particular firm. The question most often analyzed in that literature resolves around how firms' ability to discriminate along consumer brand preferences influences prices and firms' profits. Thisse and Vives (1988) were the first to show that firms end up in a prisoner's dilemma such that every firm has a unilateral incentive to discriminate, while both firms are worse-off compared to the no-discrimination case.⁸ In contrast to most articles in our model consumers differ not only in their brand preferences, but also in transport cost parameters. This allows to analyze the profit effect of price discrimination based on consumer flexibility and the resulting incentives of the firms to acquire customer flexibility data.⁹

⁶Transport cost parameter is closely related to consumer's price elasticity. All other things being equal, a consumer located farther away from a firm than from its rival is more likely to buy from it given some price discount if she is more price elastic implying a smaller transport cost parameter in a model of spatial competition.

⁷For example, since 2012 Verizon Wireless, the largest network operator in the U.S., sells data on location, demographics (age, gender) and consumption habits of its customers (Panzarino, 2012). In 2013, AT&T, the American telecommunications giant, also announced its plans to sell data on its customers' smartphone usage (see Fitchard, 2013). Similarly, Factual, a mobile marketing firm, recently launched Geopulse Audience, a data platform that allows advertisers to deliver personalized offers to consumers based not only on their locations, but also on their income, which is estimated based on their geo-behavioral patterns (see <http://www.factual.com/products/geopulse-audience>). A user's income can serve as a good proxy for her flexibility to switch between competing advertisers. Sense Networks is another mobile advertising company, which allows consumer targeting based on location and behavioral data. Information is available on age, income, education and ethnicity, all reasonable signals of consumer flexibility (see <https://www.sensenetworks.com/audience-segments-and-results/>).

⁸A similar contribution is made in Shaffer and Zhang (1995) and Bester and Petrakis (1996). Other articles show that firms' ability to discriminate along consumer locations does not necessarily lead to a prisoner's dilemma. For example, in Shaffer and Zhang (2000) firms may benefit from the ability to discriminate among the two consumer groups loyal to each of the firms if these groups are sufficiently heterogeneous in the strength of their loyalty. Chen, Narasimhan and Zhang (2001) show that when the targeting ability of one or both firms improves, but remains imperfect, firms' profits may increase. In Shaffer and Zhang (2002) a firm with a stronger brand loyalty may benefit from firms' ability to discriminate among individual consumers based on the strength of brand loyalty.

⁹Borenstein (1985) also considers a model where consumer preferences are heterogeneous along different dimensions. His simulation results show that price discrimination based on transport cost parameters is profitable. Similarly, Armstrong (2006) shows that firms benefit from the ability to discriminate between the two consumer groups with high and low transportation costs both of which are heterogeneous in brand preferences. We provide analytical results on the profit effects of firms' ability to discriminate along consumer flexibility (when they hold perfect data on consumer brand preferences). Different from the previous research we show that this effect can

Corts (1998) introduces the concepts of best-response symmetry and best-response asymmetry to predict the price (profit) effect of price discrimination. If both firms approach symmetrically the two groups of consumers among which they can price discriminate, such that the strong market of one firm is at the same time the strong market of the other, their best-response functions are characterized by best-response symmetry. In the opposite case we have best-response asymmetry. Corts shows that best-response asymmetry is a necessary condition for third-degree price discrimination to have an unambiguous effect on equilibrium prices and profits, which can be either positive or negative. As shown in the above mentioned literature, when firms discriminate along consumer brand preferences, the problem is characterized by best-response asymmetry and the profit effect of price discrimination is negative. On the other hand, a few articles analyzing the profit effect of price discrimination along consumer flexibility (Borenstein, 1985 and Armstrong, 2006) show that the problem is characterized by best-response symmetry in that case and the profit effect is positive. In the light of the results of Corts, our contribution to the literature is two-fold. First, we show that when firms hold data on consumer brand preferences, the ability to discriminate along consumer flexibility gives rise to best-response asymmetry. Then according to Corts the profit effect can be either positive or negative. Second, we show that the sign of the profit effect depends on consumer heterogeneity in flexibility and is positive if consumers are weakly differentiated and negative if consumers are strongly differentiated. To the best of our knowledge, the case with best-response asymmetry where the profit (price) effect of price discrimination is positive has not yet been considered in the literature.

The article most closely related to ours is Liu and Serfes (2004), who develop a location model of oligopolistic third-degree price discrimination to study firms' incentives to acquire data on consumer brand preferences depending on its quality. We extend their analysis by adding another dimension of consumer heterogeneity, flexibility, and allow firms to acquire data on it. Liu and Serfes show that firms acquire customer data in equilibrium only if its quality is sufficiently high. Compared to them, we identify more scope for profitable data acquisition, which at the same time harms consumers. Precisely, we show that when consumers are weakly differentiated, firms acquire customer flexibility data of any quality, which decreases consumer surplus. In contrast, when consumers are strongly differentiated, our results are similar to those of Liu and

be negative. Liu and Shuai (2013) also consider a model with two-dimensional consumer heterogeneity. However, in their analysis the strength of consumer preferences is same among all consumers.

Serfes. New results compared to Liu and Serfes are driven by the fact that in our model which is also characterized by best-response asymmetry, the profit effect of price discrimination along consumer flexibility can be either positive or negative depending on consumer flexibility, while it is always negative in Liu and Serfes.

The intuition for our results is as follows. In the absence of data on consumer flexibility the best-response function of a firm on its turf is given by either a monopolization or a market-sharing strategy depending on consumer heterogeneity in flexibility. In the former case a firm chooses to serve all consumers on its turf independently of the rival's price, which is only optimal if consumers are weakly differentiated. In contrast, when consumers are strongly differentiated, a firm prefers to serve all consumers on its turf only if the rival's price is relatively high, otherwise it serves only its more loyal consumers with high transport cost parameters. The type of a best-response function of a firm on its turf determines the intensity of price competition. Precisely, when a firm follows a monopolization strategy, the rival charges the price of zero in equilibrium. As competition is very intense initially, the acquisition of flexibility data gives rise only to a positive rent-extraction effect, and in equilibrium both firms acquire flexibility data of any quality. If a firm follows a market-sharing strategy on its turf in the absence of flexibility data, in equilibrium the rival charges there a positive price targeted at the less loyal consumers of a firm. The acquisition of flexibility data then gives rise to both a negative competition effect and a positive rent-extraction effect. In equilibrium both firms acquire flexibility data only if its quality is relatively good, in which case the latter effect is strong enough.

Our paper is also related to Jenzsch, Sapi and Suleymanova (2013). The authors show that competitors' incentives to share customer data among each other depend on its type and on how strongly consumers differ in flexibility. We show that consumer heterogeneity in flexibility is also crucial for firms' incentives to acquire customer data. Also, in our analysis customer flexibility data can be imperfect, while it is always perfect in Jenzsch, Sapi and Suleymanova.

Our paper is organized as follows. In Section 2 we present the model. In Section 3 we state the results of the equilibrium analysis. Precisely, we derive firms' equilibrium incentives to acquire customer flexibility data depending on its quality and on consumer heterogeneity in flexibility. We also derive welfare implications of firms' data acquisition decisions. In Section 4 we prove the robustness of our results and show that our main conclusions are also valid when consumers are moderately differentiated in flexibility, which allows us to make policy implications. In Section

5 we compare our results with the closely related articles of Corts (1998) and Liu and Serfes (2004). In Section 6 we discuss some of our assumptions. We analyze how firms' incentives to acquire customer flexibility data would change if data acquisition were costly. We also consider the case where firms do not hold perfect data on consumer brand preferences and are only able to distinguish among the own loyal consumers and those of the rival. Finally, in Section 7 we conclude.

2 The Model

There are two firms, A and B , producing two brands of the same product at zero marginal cost and competing in prices. Firms are situated at the two ends of a Hotelling line of unit length: Firm A is located at $x_A = 0$ and firm B at $x_B = 1$. There is a unit mass of consumers, each of whom is characterized by an address $x \in [0, 1]$, which corresponds to her preference for the ideal product. If a consumer does not buy her ideal product she incurs linear transport costs proportional to the distance to the firm. We follow Jentzsch, Sapi and Suleymanova (2013) and assume that additionally to their addresses consumers are also differentiated in transport costs per unit distance, $t \in [\underline{t}, \bar{t}]$, where $\bar{t} > \underline{t} \geq 0$. Each consumer is uniquely characterized by a pair (x, t) . With t and x being uniformly and independently distributed, we have the density functions: $f_t = 1/(1 - \underline{t})$, $f_x = 1$ and $f_{t,x} = 1/(1 - \underline{t})$. We will refer to consumers with $x < 1/2$ ($x > 1/2$) as the *turf* of firm A (B). We will also distinguish among consumers with the same address and will refer to consumers with a larger (smaller) transport cost parameter as *more* (*less*) *loyal*.

In the main analysis we consider two extreme versions of our model, depending on the level of consumer heterogeneity in flexibility measured by the ratio of the largest to the smallest transport cost parameters, $l := \bar{t}/\underline{t}$. In the first version $\underline{t} = 0$, such that $\lim_{\underline{t} \rightarrow 0} l = \infty$. We say that in this case consumers are *strongly differentiated in flexibility*. In the second version $\underline{t} > 0$ and $l \leq 2$ and consumers are *weakly differentiated in flexibility*. Since all the results in our model are driven by the ratio l , without loss of generality we normalize $\bar{t} = 1$.

We assume that firms have perfect data on consumer addresses and, hence, can practice (perfect) third-degree price discrimination along that dimension. Additionally, firms can acquire imperfect data on consumer flexibility, which allows them to practice (imperfect) third-degree price discrimination along consumer flexibility. In particular, we assume that firms can acquire

for free an external dataset containing the flexibility characteristics of consumers in the market.¹⁰ If firms acquire this dataset, they can identify the transport cost parameters of individual consumers, or of consumer groups, depending on data quality. In particular, the quality of data on consumer flexibility is measured by parameter $k = 0, 1, 2, \dots, \infty$. For a given k firms can divide the interval $t \in [\underline{t}, 1]$ into 2^k segments and identify every consumer as belonging to one of those segments.¹¹ Segment $m = 1, 2, \dots, 2^k$ consists of consumers with transport cost parameters $t \in [\underline{t}^m(k); \bar{t}^m(k)]$, where $\underline{t}^m(k) = \underline{t} + (1 - \underline{t})(m - 1)/2^k$ and $\bar{t}^m(k) = \underline{t} + (1 - \underline{t})m/2^k$ denote the most and the least flexible consumers on segment m , respectively. For any m we can compute the ratio of the largest to the lowest transport cost parameters on that segment, $l^m(k) := \bar{t}^m(k)/\underline{t}^m(k)$. We say that consumers are *strongly differentiated in flexibility on segment m* if $\underline{t}^m(k) = 0$, such that $\lim_{\underline{t}^m(k) \rightarrow 0} l^m(k) = \infty$. Similarly, consumers are *weakly differentiated in flexibility on segment m* if $\underline{t}^m(k) > 0$ and $l^m(k) \leq 2$. Note that segment $m = 1$ always contains the most flexible consumer (with $t = \underline{t}$). We will show in the next section that this is the only segment where consumers can be strongly differentiated (and only in a version of our model with strongly differentiated consumers).

When $k \rightarrow \infty$, firms have perfect data on the addresses and flexibility of all consumers in the market and, hence, can charge individual prices. In all other cases firms have to charge the same price to consumers on one segment with the same address. We denote the price of firm i to consumers with address x on segment m when customer data is of quality k as $p_{im}(x, k)$.

The utility of a consumer (x, t) from buying at firm $i = \{A, B\}$ is

$$U_i(p_{im}(x, k), t, x) = v - t|x - x_i| - p_{im}(x, k). \quad (1)$$

In equation (1) $v > 0$ denotes the basic utility, which is high enough such that the market is always covered in equilibrium. A consumer buys from a firm proposing a higher utility. We follow Thisse and Vives (1988) and assume that if a consumer is indifferent, she buys from the closer firm. If $x = 1/2$, then in the case of indifference a consumer buys from firm A . The game

¹⁰The assumption that firms can acquire customer flexibility data for free allows us to concentrate on the profit effects of price discrimination based on consumer flexibility. In Section 6 we analyze how our results on firms' incentives to acquire customer data would change if data acquisition were costly.

¹¹As we mentioned in the Introduction, in the marketing literature it is common to model consumer heterogeneity along two dimensions: brand preferences and responsiveness to marketing variables, such as price and advertising. For example, based on a sample of weekly store-level data for ketchup, Besanko, Dubé and Gupta (2003) identify three flexibility segments characterized by different price elasticities: price-insensitive, moderately price-sensitive and very price-sensitive shoppers.

unfolds as follows.

Stage 1 (Customer data acquisition). Firms observe the exogenously given quality of customer flexibility data, k , and decide independently from each other whether to acquire this data.

Stage 2 (Competition). First, firms independently and simultaneously choose prices for each address x . Subsequently the firm(s) with customer flexibility data issues (issue) discounts to consumers on different flexibility segments.

The timing of the competition stage is consistent with a large body of literature on competitive price discrimination where firms make their targeted offers after setting regular prices (e.g., Thisse and Vives, 1988; Shaffer and Zhang, 1995, 2002; Liu and Serfes, 2004, 2005).¹² It reflects the observation that discounts issued to finer consumer groups can be changed easier than prices targeted at broader consumer groups.¹³ Moreover, if firms decide simultaneously on regular prices and discounts, Nash equilibrium in pure strategies does not always exist.

3 Equilibrium Analysis

We solve the game backwards and start with the competition stage where firms choose prices taking their decisions in the data acquisition stage as given. Two subgames can emerge in the second stage. In the *symmetric subgame* both firms hold customer flexibility data.¹⁴ In the *asymmetric subgame* only one firm holds data on consumer flexibility. We derive the equilibrium in each subgame and compare profits in different subgames to conclude on firms' incentives to acquire customer flexibility data in the first stage. We denote the equilibrium profit of firm $i = A, B$ as $\Pi_i^{D,D}(k)$ ($\Pi_i^{D,ND}(k)$) in the symmetric (asymmetric) subgame when data quality is given by k .

Before we proceed with the main analysis, we derive some results on the segment types in each version of our model depending on data quality. The following lemma summarizes our results.

¹²Note that the timing in Stage 2 is equivalent to the following: *i*) in the subgame where both firms hold flexibility data, firms choose all the prices simultaneously, and *ii*) in the subgames where only one firm holds flexibility data, the firm without data chooses its prices first, and the other firm follows.

¹³We observe in many markets that regular prices change less frequently than coupon discounts. For example, www.hutfans.com (retrieved on January 26, 2013) mentions that Pizza Hut discount coupons tend to "change often."

¹⁴If $k = 0$, this subgame is equivalent to neither of the firms holding customer flexibility data.

Lemma 1. (Segment types.) *In the version of our model with weakly differentiated consumers, for any data quality $k \geq 1$ consumers are weakly differentiated on any segment. In the version of our model with strongly differentiated consumers, for any data quality $k \geq 1$ consumers are weakly differentiated on any segment $m \geq 2$ and are strongly differentiated on segment $m = 1$.*

Proof. See Appendix.

If consumers are weakly differentiated, with data of a better quality they are divided into flexibility segments, where they become even more homogeneous, such that on all segments consumers remain weakly differentiated. If consumers are strongly differentiated, under any data quality only on segment $m = 1$ there is a consumer with a zero transport cost parameter, such that consumers remain strongly differentiated only on that segment. On all other segments consumers become weakly differentiated.

3.1 Symmetric Subgame: Both Firms Hold Data on Consumer Flexibility

When both firms hold data on consumer flexibility, they can identify each consumer as belonging to one of the flexibility segments and can charge different prices on different segments. As firms are symmetric, we will focus on the turf of firm A . Consider some $x < 1/2$ and an arbitrary m . Given prices $p_{Am}(x)$ and $p_{Bm}(x)$, the transport cost parameter of the indifferent consumer on segment m with address x is

$$\tilde{t}^m(x; p_{Am}(x), p_{Bm}(x)) = \frac{p_{Am}(x) - p_{Bm}(x)}{1 - 2x}, \text{ provided } \tilde{t}_m(\cdot) \in [\underline{t}^m(k); \bar{t}^m(k)].$$

On segment m firm A serves consumers with high transport cost parameters, $t \geq \tilde{t}_m(\cdot)$. Firm B attracts consumers with low transport cost parameters, $t < \tilde{t}_m(\cdot)$. Then on any address $x < 1/2$ and segment m under data quality k firm A maximizes the expected profit

$$\Pi_{Am}(x; p_{Am}(x), p_{Bm}(x)) := p_{Am}(x) \Pr \{t \geq \tilde{t}^m(\cdot)\}$$

by choosing the price function $p_{Am}(x)$. Firm B maximizes the expected profit

$$\Pi_{Bm}(x; p_{Am}(x), p_{Bm}(x)) := p_{Bm}(x) \Pr \{t < \tilde{t}^m(\cdot)\}$$

by choosing the price function $p_{Bm}(x)$.

As we showed in Lemma 1, in both versions of our model under any data quality a flexibility segment can be of two types: with weakly or strongly differentiated consumers. It is then useful to derive first the equilibrium on a flexibility segment of each type. As consumers can be strongly differentiated only on segment $m = 1$, we consider $\underline{t}^1(k) = 0$ (for any k) in that case. The following lemma summarizes our results.

Lemma 2. (Equilibria on different flexibility segments.) *Consider some x on the turf of firm $i = \{A, B\}$ and some m . Assume that both firms hold customer flexibility data of quality k . Equilibrium prices, demand regions and profits on x and m depend on consumer heterogeneity.*

i) Assume that consumers are weakly differentiated on m . Then on x and m firms charge prices $p_{im}^(x, k) = \underline{t}^m(k) |1 - 2x|$ and $p_{jm}^*(x, k) = 0$, where firm i serves all consumers and firms realize profits $\Pi_{im}(x, k) = \underline{t}^m(k) |1 - 2x|$ and $\Pi_{jm}(x, k) = 0$.*

ii) Assume that consumers are strongly differentiated on $m = 1$. Then on x and $m = 1$ firms charge prices $p_{im}^(x, k) = 2 |1 - 2x| / (3 \times 2^k)$ and $p_{jm}^*(x, k) = |1 - 2x| / (3 \times 2^k)$, where firm i serves consumers with $t \geq 1 / (3 \times 2^k)$ and firms realize profits $\Pi_{im}(x, k) = 4 |1 - 2x| / (3 \times 2^k)^2$ and $\Pi_{jm}(x, k) = |1 - 2x| / (3 \times 2^k)^2$.*

Proof. See Appendix.

As Lemma 2 shows, we get different equilibria on a flexibility segment depending on its type. On a segment with weakly differentiated consumers a firm serves all its loyal consumers. On a segment with strongly differentiated consumers a firm loses the more flexible consumers to the rival. This difference in equilibria is driven by the type of a best-response function of a firm on its turf. Precisely, on any address on its turf a firm follows a *monopolization strategy* on segments with weakly differentiated consumers and on segments with strongly differentiated consumers a firm resorts to a *market-sharing strategy*. To demonstrate this we turn to the turf of firm A . Consider first an arbitrary segment m with weakly differentiated consumers, where $\underline{t}^m(k) \leq 2$ holds. The best-response function of firm A on m on some $x < 1/2$ is

$$p_{Am}(x, k, p_{Bm}) = p_{Bm} + \underline{t}^m(k)(1 - 2x), \text{ for any } p_{Bm}. \quad (2)$$

As expression (2) shows, for any price of the rival firm A optimally charges a relatively low price targeted at the most flexible consumer on segment m (with $t = \underline{t}^m(k)$), which allows to

serve all consumers there. We say that firm A follows a monopolization strategy in that case. Why is such a strategy optimal on a segment with weakly differentiated consumers? The reason is that it suffices for firm A to reduce slightly its price targeted at the less flexible consumers on segment m to gain all consumers there, which makes it optimal for firm A to monopolize segment m independently of the rival's price. Then in equilibrium firm B charges the price of zero on segment m and does not serve any consumers there.

Consider now segment m with strongly differentiated consumers, where $\lim_{\underline{t}^m(k) \rightarrow 0} \bar{t}^m(k) = \infty$ holds. The best-response function of firm A on m on some $x < 1/2$ is

$$p_{Am}(x, k, p_{Bm}) = \begin{cases} p_{Bm} & \text{if } p_{Bm} \geq \bar{t}^m(k)(1 - 2x) \\ [p_{Bm} + \bar{t}^m(k)(1 - 2x)] / 2 & \text{if } p_{Bm} < \bar{t}^m(k)(1 - 2x). \end{cases} \quad (3)$$

Since the most flexible consumer on segment m can switch brands costlessly ($\underline{t}^m(k) = 0$), in order for firm A to attract all consumers there it has to charge a price, which is at least as low as that of the rival (as the first line in (3) shows). This is only optimal for firm A if the rival's price is sufficiently high ($p_{Bm} \geq \bar{t}^m(k)(1 - 2x)$). Otherwise, firm A charges a higher price targeted at the less flexible consumers ($p_{Am}(\cdot) = [p_{Bm} + \bar{t}^m(k)(1 - 2x)] / 2$) and lets the rival gain the more flexible ones. We say that firm A follows a *market-sharing strategy*: It optimally shares segment m with the rival when serving all consumers there is too costly, which is the case if the rival's price is relatively low ($p_{Bm} < \bar{t}^m(k)(1 - 2x)$). As a result, in equilibrium both firms charge positive prices and serve consumers on segment m . Lemmas 1 and 2 state equilibrium prices, demand regions and profits on any address and any segment separately in both versions of our model. In the next proposition we summarize those results and state the equilibrium in each version of our model depending on data quality.

Proposition 1. (Symmetric subgame.) *Assume that both firms hold customer flexibility data of quality k . Equilibrium prices, demand regions and firms' profits depend on how strongly consumers differ in flexibility.*

i) *If consumers are weakly differentiated, then on the turf of firm i on any segment firms charges prices $p_{im}^*(x, k) = \underline{t}^m(k) |1 - 2x|$ and $p_{jm}^*(x, k) = 0$ and realize profits $\Pi_i^{D,D}(k) = (1 + \underline{t}) / 8 - (1 - \underline{t}) / 2^{3+k}$. Firm i serves all consumers on its turf.*

ii) *If consumers are strongly differentiated, then on the turf of firm i on segment $m = 1$ firms charges prices $p_{im}^*(x, k) = 2 |1 - 2x| / (3 \times 2^k)$ and $p_{jm}^*(x, k) = |1 - 2x| / (3 \times 2^k)$, where firm*

i serves consumers with $t \geq 1/(3 \times 2^k)$. On segments $m \geq 2$ firms charge prices $p_{im}^*(x, k) = \underline{t}^m(k) |1 - 2x|$ and $p_{jm}^*(x, k) = 0$, where firm i serves all consumers. Firms realize profits $\Pi_i^{D,D}(k) = (10 - 9 \times 2^k) / (9 \times 2^{3+2k}) + 1/8$.

Proof. See Appendix.

When $k = 0$, firms can only discriminate along consumer brand preferences. With weakly differentiated consumers every firm follows a monopolization strategy on any address on its turf and serves all consumers there. With strongly differentiated consumers a firm follows a market-sharing strategy and serves only the less flexible consumers on its turf. Figure 1 shows equilibrium demand regions in both versions of our model for $k = 0$.

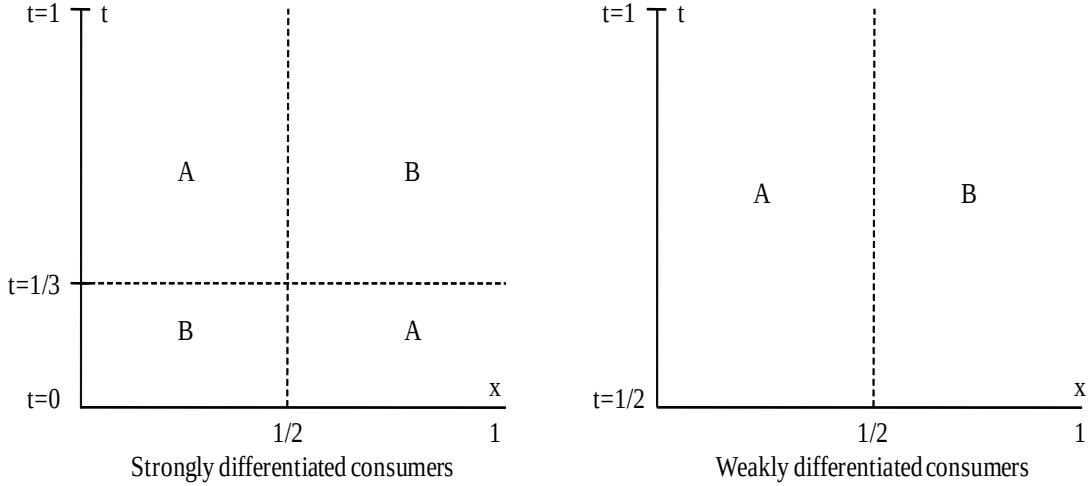


Figure 1: Equilibrium demand regions for $k = 0$. We used $\underline{t} = 1/2$ in the case of weakly differentiated consumers.

How do firms' profits change when the quality of customer flexibility data improves and takes values $k \geq 1$? In the literature on competitive price discrimination one usually distinguishes between two effects. Data of a better quality potentially allows firms to extract more rents from consumers, to which we refer as the rent-extraction effect. It may also change the intensity of price competition, to which we refer as competition effect.

Consider first the case of weakly differentiated consumers. As shown in Lemma 1, for any data quality consumers remain weakly differentiated on any segment such that every firm resorts to a monopolization strategy on its turf for any data quality and serves all consumers there. In equilibrium each firm targets the most flexible consumer on a given segment on any address on

its turf while the rival charges the price to zero. Figure 2 shows equilibrium demand regions in both versions of our model for $k = 1$. The improvement in customer data by one step results in a situation, in which every consumer is charged a (weakly) higher price. This is because each consumer is allocated to a segment in which the most flexible consumer is (weakly) less flexible than before. As a result firms' profits unambiguously increase. Higher quality data improves firms' ability to extract rents from their loyal consumers, while competition effect is absent as the rival cannot go below the price of zero on a firm's turf. Figure 3 depicts a firm's equilibrium profit as a function of customer data quality in the symmetric subgame in the two versions of our model.¹⁵

Consider now the case of strongly differentiated consumers. As we showed in Lemma 1, for any quality of customer data consumers remain strongly differentiated only on segment $m = 1$, while on all other segments they become weakly differentiated. If $k = 0$, firms pursue a market-sharing strategy on their turfs. However, as k increases by one step, on segment $m = 2$ firms switch to a monopolization strategy while on segment $m = 1$ they maintain a market-sharing strategy. As a result, on both segments on any address on its turf a firm charges lower prices compared to the uniform price at $k = 0$. The reason is that on segment $m = 1$ consumers become on average more flexible, and on segment $m = 2$ the rival responds aggressively to a firm's monopolization strategy and decreases its price to zero. Lower prices on a firm's turf result in lower equilibrium profits. However, when k increases further, profits start to increase. To understand this result consider some $k \geq 1$. On (all) segment(s) $m \geq 2$, where consumers are weakly differentiated, a firm's profits on its turf increase when k gets larger by one step, along the logic explained above for the case of weakly differentiated consumers. On segment $m = 1$, where consumers are strongly differentiated, a firm's profits decrease due to the logic described above for the case $k = 0$. As segment $m = 1$ comprises (weakly) less than half of all consumers on a firm's turf for any $k \geq 1$, the negative profit effect on that segment is outweighed by the

¹⁵While Figure 3 shows how firms' profits change with the improvement in data quality for a given $\underline{t}$, it is also interesting to analyze how firms' profits change in $\underline{t}$ for a given data quality when consumers are weakly differentiated. As the derivative of the profit function with respect to $\underline{t}$ is positive for any k , we conclude that firms' profits increase when consumers become more homogeneous (and data quality is kept fixed). This result is quite intuitive. Indeed, firms' profits in the symmetric subgame with weakly differentiated consumers depend only on their ability to extract rents from consumers (competition effect is irrelevant in that case). When consumers become less differentiated ($\underline{t}$ increases), for a given data quality, on any segment the price paid by consumers becomes closer to the price, which makes them indifferent between buying at the two firms, which implies that firms extract more rents from consumers. We can state this result differently. When consumers become more homogeneous, the average transport cost parameter on a given segment conveys more information about the individual transport cost parameter of each consumer, which is equivalent to an increase in data quality.

positive profit effect on all the other segment(s), such that a firm's profits on its turf increase. For the same reason, with an increase in k each firm serves more consumers on its turf as for any $k \geq 1$ firms lose consumers on their turfs only on segment $m = 1$ (compare Figures 1 and 2).

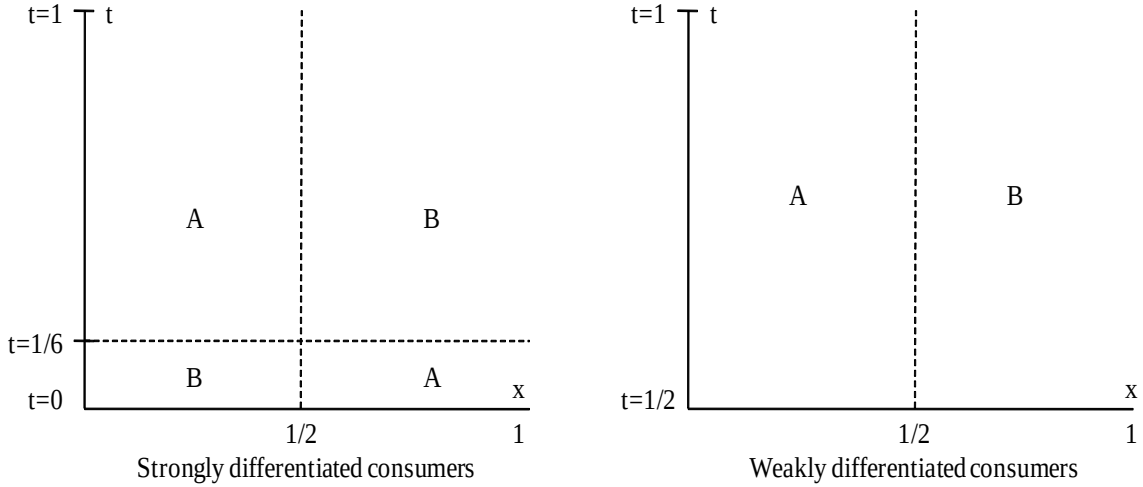


Figure 2: Equilibrium demand regions for $k = 1$. We used $\underline{t} = 1/2$ in the case of weakly differentiated consumers.

We can summarize our results as follows. When consumers are weakly differentiated, in the absence of flexibility data each firm follows a monopolization strategy on its turf, such that the rival charges the price of zero there and competition is very intense. The improvement in data quality results then only in a positive rent-extraction effect, while competition cannot be intensified as the rival cannot go below the price of zero. When consumers are strongly differentiated, in the absence of flexibility data each firm follows a market-sharing strategy on its turf, where the rival charges positive prices and competition is not very intense. Then the improvement in data quality gives rise to both a negative competition effect and a positive rent-extraction effect. The latter becomes strong enough when firms hold data of a high quality. As a result, when data quality improves from $k = 0$ to $k = 1$, profits decrease and start to increase afterwards. However, they never reach the initial level at $k = 0$ and the negative competition effect dominates with data of any quality.

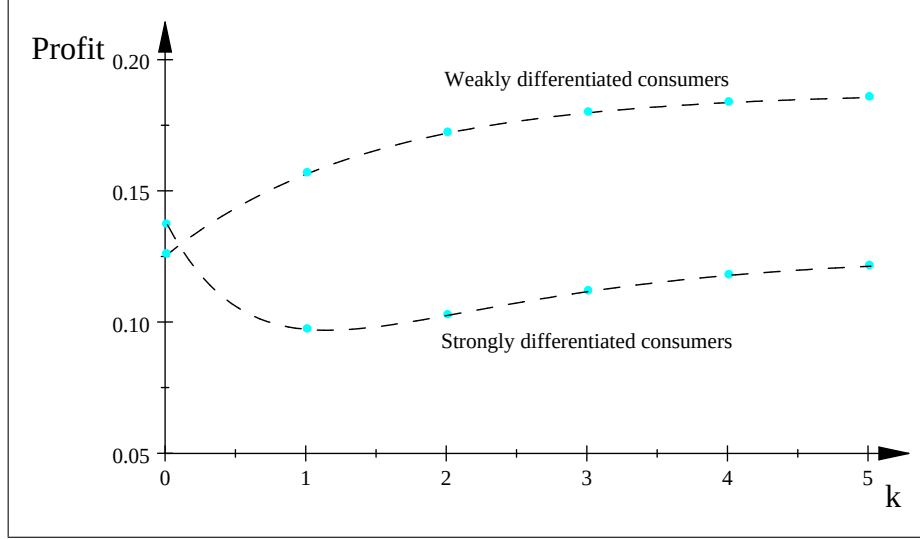


Figure 3: Individual profits and the quality of customer flexibility data (symmetric subgame). We used $\underline{t} = 1/2$ in the case of weakly differentiated consumers.

3.2 Asymmetric Subgame: Only One Firm Holds Data on Consumer Flexibility

We assume without loss of generality that firm A acquires customer flexibility data while firm B does not. The latter must offer the same price to all consumers with a given address irrespective of their flexibility. In contrast, firm A can price-discriminate based on both consumer addresses and flexibility. The following proposition summarizes our results for the case of weakly differentiated consumers.

Proposition 2. (Asymmetric subgame with weakly differentiated consumers.) *Assume that consumers are weakly differentiated and only firm A holds data on consumer flexibility. Equilibrium prices, demand regions and profits depend on consumer flexibility and the quality of customer data.*

i) *If $l \leq 3/2$, then for any $k \geq 0$ on the turf of firm A firms charge prices $p_{Am}^*(x, k) = \underline{t} [1 + (l - 1)(m - 1)/2^k] (1 - 2x)$ and $p_B^*(x, k) = 0$. On the turf of firm B firms charge prices $p_{Am}^*(x, k) = 0$ and $p_B^*(x, k) = \underline{t}(2x - 1)$. Every firm serves all consumers on its turf. Firms realize profits $\Pi_A^{D,ND}(k) = \underline{t}(1 + l)/8 - \underline{t}(l - 1)/2^{k+3}$ and $\Pi_B^{D,ND}(k) = \underline{t}/4$.*

ii) *If $3/2 < l \leq 2$ and $0 \leq k < \log_2 [2(l - 1)/(2l - 3)]$, then on the turf of firm A firms charge prices $p_{Am}^*(x, k) = \underline{t} [1 + (l - 1)(m - 1)/2^k] (1 - 2x)$ and $p_B^*(x, k) = 0$, where firm A serves all consumers. On the turf of firm B firms charge prices $p_{A1}^*(x, k) = \underline{t}(2l - 3)(2x - 1)/4$*

and $p_{Am}^*(x, k) = 0$ if $m > 1$ and $p_B^*(x, k) = \underline{t}(2l - 1)(2x - 1)/2$, where firm A serves consumers with $t < \underline{t}(2l + 1)/4$. Firms realize profits $\Pi_A^{D,ND}(k) = \underline{t}[12l^2 - 12l + 1]/[64(l - 1)] - \underline{t}(l - 1)/2^{k+3}$ and $\Pi_B^{D,ND}(k) = \underline{t}(2l - 1)^2/[32(l - 1)]$.

iii) If $3/2 < l \leq 2$ and $k \geq \log_2[2(l - 1)/(2l - 3)]$, then on the turf of firm A firms charge prices $p_{Am}^*(x, k) = \underline{t}[1 + (l - 1)(m - 1)/2^k](1 - 2x)$ and $p_B^*(x, k) = 0$, where firm A serves all consumers. On the turf of firm B firms charge prices $p_{A1}^*(x, k) = \underline{t}(l - 1)(2x - 1)/2^{k+1}$ and $p_{Am}^*(x, k) = 0$ if $m > 1$ and $p_B^*(x, k) = \underline{t}[1 + (l - 1)/2^k](2x - 1)$, where firm A serves consumers with $t < \underline{t}[1 + (l - 1)/2^{k+1}]$. Firms realize profits $\Pi_A^{D,ND}(k) = \underline{t}(l + 1)/8 - \underline{t}(l - 1)(2^{k+1} - 1)/2^{2k+4}$ and $\Pi_B^{D,ND}(k) = \underline{t}[1 + (l - 1)/2^k](1 - 1/2^{k+1})/4$.

Proof. See Appendix.

For the intuition behind the equilibrium consider first the turf of firm A . For any data quality consumers remain weakly differentiated on any flexibility segment, where firm A adopts a monopolization strategy and targets the most flexible consumer. Firm B cannot do better than charging the price of zero on any address for any data quality. Therefore firm A does not face a negative competition effect when data quality improves. In the latter case every consumer is allocated to a segment in which the most flexible consumer is now (weakly) less flexible than before. As a result, the profits of firm A on its own turf increase with data quality improvement due to a positive rent-extraction effect. Firm B does not serve any consumers on firm A 's turf and does not earn any profits there.

Consider now the turf of firm B . Different from the symmetric subgame where in equilibrium each firm serves all consumers on its own turf, in the asymmetric subgame the firm without flexibility data loses some of its loyal consumers if $l > 3/2$. To understand this result, assume that firm B targets the most flexible consumers on its turf and charges $p_B(x, k) = \underline{t}(2x - 1)$ for any $x > 1/2$. Does firm B have an incentive to increase prices and lose some of the more flexible consumers (on segment $m = 1$)? Compared to the symmetric subgame, firm B loses less consumers in case of a price increase because firm A responds with a higher price on segment $m = 1$. Different from the symmetric subgame, this makes it profitable for firm B to charge a price above $p_B(x, k) = \underline{t}(2x - 1)$ and lose some consumers on segment $m = 1$ if $l > 3/2$ holds.

How does the equilibrium change with the improvement in data quality? If consumers are quite homogeneous ($l \leq 3/2$), irrespectively of data quality firm B targets the most flexible consumer on any address on its turf. This leaves no scope for firm A to attract away any of its

loyal consumers, even with perfect customer data. Then both firms' profits on firm B 's turf do not change with the improvement in data quality.

Consider now the case with $l > 3/2$. If data quality is low ($k < \log_2 [(l-1)/(l-3/2)]$), firm A has a limited ability to attract consumers on the turf of firm B . In that case firm B does not have to reduce its prices with an increase in k to keep consumers on segments $m \geq 2$. Then firms' profits on the turf of firm B do not depend on k . However, when k increases above $k = \log_2 [(l-1)/(l-3/2)]$, the improved ability of firm A to target consumers forces firm B to reduce its prices to avoid losing consumers on segments $m \geq 2$. Although due to a decrease in firm B 's equilibrium prices it gains consumers on its own turf, its profits there decrease. The profits of firm A on the turf of firm B decrease as well as it has to reduce its prices on any address on segment $m = 1$, while its market share gets also smaller. As the profits of firm A on the rival's turf constitute only a small share of its total profits, the latter undoubtedly increase with the improvement in data quality. Figures 4a and 4b depict firms' profits as a function of data quality when consumers are weakly differentiated in the cases with $l \leq 3/2$ and $l > 3/2$, respectively.

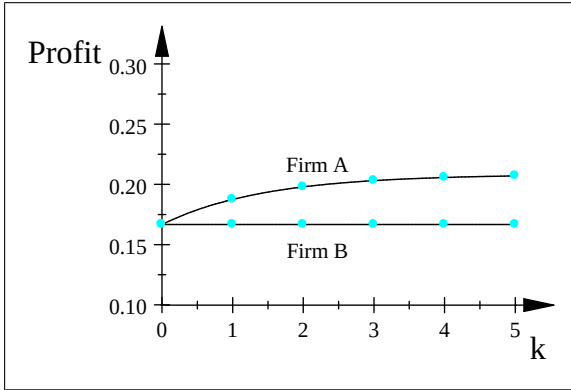


Figure 4a: $l \leq 3/2$, with $\bar{t} = 2/3$.

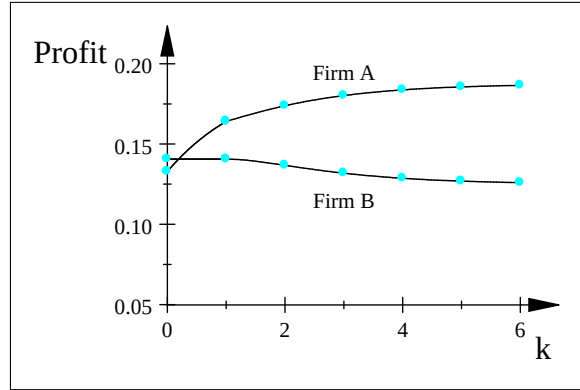


Figure 4b: $3/2 < l \leq 2$, with $\bar{t} = 1/2$.

Figures 4a-4b: Individual profits and the quality of customer flexibility data (asymmetric subgame, weakly differentiated consumers).

We can summarize our results as follows. Firm A follows a monopolization strategy on its turf, which leaves no scope for a negative competition effect with data quality improvement as the rival always charges the price of zero on firm A 's turf. Then customer data of a better quality allows firm A to extract more rents on its own turf where its profits increase in k . Firm B serves all consumers on its turf only when they are quite homogeneous ($l \leq 3/2$), in which

case again the negative competition effect is absent, and the profits of firm B do not change with data quality. Otherwise, firm B loses some consumers on its turf and suffers from the intensified competition when the rival gets data of a better quality, such that its profits decrease when k gets larger. We now turn to the case of strongly differentiated consumers. The following proposition states the equilibrium in the asymmetric subgame.

Proposition 3. (Asymmetric subgame with strongly differentiated consumers.) *Assume that consumers are strongly differentiated in flexibility and only firm A holds data on consumer flexibility. Equilibrium prices, demand regions and profits depend on the quality of customer data.*

- i) If $k = 0$, then on the turf of firm A firms charge prices $p_{Am}^*(x, k) = 3(1 - 2x)/4$ and $p_{Bm}^*(x) = (1 - 2x)/2$, where firm A serves consumers with $t \geq 1/4$. On the turf of firm B firms charge prices $p_{Am}^*(x, k) = (2x - 1)/2$ and $p_{Bm}^*(x) = (2x - 1)$, where firm A serves consumers with $t < 1/2$. Firms realize profits $\Pi_A^{D,ND}(k) = 13/64$ and $\Pi_B^{D,ND}(k) = 5/32$.*
- ii) If $k \geq 1$, then on the turf of firm A firms charge prices $p_{A1}^*(x, k) = 3(1 - 2x)/2^{k+2}$ and $p_{Am}^*(x, k) = \underline{t}^m(k)(1 - 2x) + p_B^*(x)$ if $m > 1$ and $p_B^*(x) = (1 - 2x)/2^{k+1}$, where firm A serves consumers with $t \geq 1/2^{k+2}$. On the turf of firm B firm A charges prices $p_{Am}^*(x, k) = p_B^*(x) - \bar{t}^m(k)(2x - 1)$ if $m < 2^{k-1}$, $p_{A,2^{k-1}}^*(x, k) = 5(2x - 1)/2^{k+3}$, $p_{A,2^{k-1}+1}^*(x, k) = (2x - 1)/2^{k+3}$ and $p_{Am}^*(x, k) = 0$ if $m > 2^{k-1} + 1$. On its own turf firm B charges the price $p_B^*(x) = (2x - 1)(1 + 1/2^{k+1})/2$, where firm A serves consumers with $t < (2^k/2 - 3/8)/2^k$ and $1/2 < t < (2^k/2 + 1/8)/2^k$. Profits are $\Pi_A^{D,ND}(k) = (5 - 1/2^k + 7/2^{2k+2})/32$ and $\Pi_B^{D,ND}(k) = (1 + 1/2^k + 1/2^{2k+2})/16$.*

Proof. See Appendix.

Although firm A has an informational advantage over the rival, it does not serve all consumers on its turf. This happens for the same reason as in the symmetric subgame. Since on segment $m = 1$ consumers always remain strongly differentiated, firm A follows a market-sharing strategy there and loses the more flexible consumers. Due to a negative competition effect the profits of firm A on its turf decrease when its targeting ability improves. Precisely, anticipating a stronger informational advantage of firm A following an increase in data quality, firm B targets on each address on firm A 's turf more flexible consumers and has to charge lower prices to them. Those prices in turn serve as an anchor for the discriminatory prices of firm A on each address and push them downwards. As a result, under any data quality firm A faces a strong negative competition

effect on its turf, such that its profits there decrease monotonically in k .¹⁶ The profits of firm B on the rival's turf decrease as well, because both its price and the market share become smaller.

Due to its informational advantage in the asymmetric subgame firm A gains about half of all consumers on the turf of firm B , whereas in the symmetric subgame it serves only consumers on segment $m = 1$. The profits of firm A on the rival's turf exhibit a U -shaped relationship in k : They first decrease and start to increase at $k = 1$. Firm B reduces the uniform prices on its turf to protect its market shares, which constitutes a negative competition effect for firm A 's profits there. However, different from the turf of firm A , where on each address firm B targets the more flexible consumers, on its own turf firm B targets the less flexible consumers. Then although the prices of firm B on each address on its turf decrease, they approach some positive values and not the price of zero as on the turf of firm A . As a result, the strength of the negative competition effect diminishes with the improvement in data quality, and the profits of firm A on the rival's turf start to increase at $k = 1$ as firm A is able to extract more rents from consumers it serves there. The profits of firm B earned on the own turf decrease monotonically with the improvement in data quality. It is forced into a downward spiral where it must charge lower prices while it still loses market shares.

Figure 5 shows the combined effect of data quality on firms' profits on the two turfs. The profits of firm B decrease monotonically with the improvement in data quality. The profits of firm A exhibit a U -shaped relationship in data quality, exactly as in the symmetric subgame (compare with Figure 3).

¹⁶Note that in the symmetric subgame with strongly differentiated consumers the profits of firm A on its turf exhibit a U -shaped relationship in k . In that case the strength of the negative competition effect for firm A 's profits is limited. As firm B can discriminate, it reduces its price to zero on segments with relatively homogeneous consumers ($m \geq 2$), where firm A follows a monopolization strategy. Then on those segments data improvement gives rise only to a positive rent-extraction effect, where the profits of firm A increase. Competition is only intensified on segment $m = 1$, where the profits of firm A constitute a decreasing share of its total profits on the own turf when data quality improves.

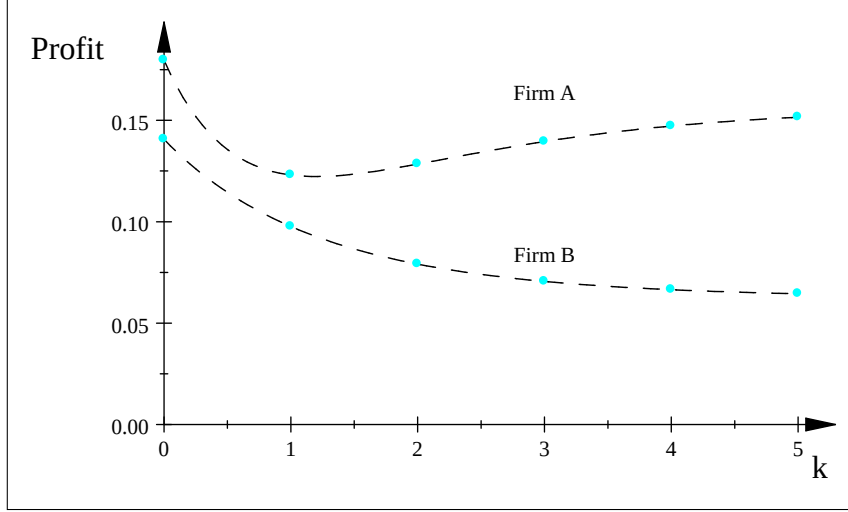


Figure 5: Individual profits and the quality of customer flexibility data (asymmetric subgame, strongly differentiated consumers).

3.3 Acquisition of Customer Data

In this subsection we analyze firms' incentives to acquire customer flexibility data in the first stage of the game and its welfare implications. In particular, we assume that firms can obtain data on the flexibility of all consumers in the market with an exogenously given precision $k \geq 1$. To concentrate on the profit effects of data acquisition we assume that customer data can be acquired for free. The following proposition summarizes the main results depending on consumer heterogeneity in flexibility.

Proposition 4. (Customer-data acquisition.) *Firms' decisions to acquire customer data depend on how strongly consumers differ in flexibility.*

- i) *When consumers are weakly differentiated, for any $k \geq 1$ there is a unique Nash equilibrium (in dominant strategies) where both firms acquire customer flexibility data.*
- ii) *When consumers are strongly differentiated, there are two Nash equilibria where only one of the firms acquires customer data if $k = 1$. If $k \geq 2$ there is a unique Nash equilibrium (in dominant strategies) where both firms acquire customer flexibility data.*

Proof. See Appendix.

With weakly differentiated consumers in the absence of customer flexibility data every firm serves all consumers on its own turf and targets on each address the most flexible consumer.

If one of the firms acquires customer data, it gains consumers on the rival's turf (provided consumers are not very homogeneous, $l > 3/2$) and extracts more rents from consumers on the own turf. The unilateral acquisition of customer data is then always profitable. The best response to the rival acquiring customer flexibility data is to do so as well. In that case a firm gains consumers on the own turf (provided consumers are not very homogeneous, $l > 3/2$) and extracts more rents from them. In equilibrium both firms acquire data of any quality.

With strongly differentiated consumers every firm has a unilateral incentive to acquire customer data of any quality. Although firm B (the firm without data) responds to data acquisition by charging lower prices on both the rival's and the own turf, unilateral data acquisition is still profitable because firm A gains consumers on both turfs due to its improved targeting ability. In contrast, acquiring data when the rival holds it is only profitable if data quality is sufficiently high ($k \geq 2$). On the one hand, data acquisition allows firm B to gain consumers on its own turf. On the other hand, on those segments on firm B 's turf where firm A loses consumers, it reduces its price to zero. Only when data quality is high enough, is this negative effect compensated by the improved ability of firm B to extract rents from the less flexible consumers on its turf (to whom firm A always charges the price of zero independently of whether firm B holds flexibility data). These data acquisition incentives result in the equilibrium where both firms acquire flexibility data only when its quality is sufficiently high ($k \geq 2$). Otherwise (if $k = 1$), only one firm acquires customer data in equilibrium. In the next proposition we characterize equilibrium profits, consumer surplus and social welfare under any data quality based on firms' equilibrium decisions to acquire customer data in the first stage of the game.

Proposition 5. (Profit and welfare depending on data quality.) *Firms' profits, consumer surplus and social welfare depend on consumer heterogeneity in flexibility and the quality of customer data.*

i) If consumers are weakly differentiated, under any $k \geq 0$ the profits of firm $i = \{A, B\}$, consumer surplus and social welfare are $\Pi_i^(k) = \underline{t} [1/4 + (l-1)(1/8 - 1/2^{3+k})]$, $CS(k) = v - 3\underline{t}(l+1)/8 + \underline{t}(l-1)/2^{k+2}$ and $SW = v - \underline{t}(l+1)/8$, respectively.*

ii) If consumers are strongly differentiated, with $k = 0$ firms' profits, consumer surplus and social welfare are $\Pi_A^ = \Pi_B^* = 5/36$, $CS = v - 31/72$ and $SW = v - 11/72$, respectively. If $k = 1$, then $\Pi_A^* = 79/512$, $\Pi_B^* = 27/256$, $CS = v - 417/1024$ and $SW = v - 151/1024$. If $k \geq 2$, the profits of firm $i = \{A, B\}$, consumer surplus and social welfare are given by*

$\Pi_i^*(k) = 5 / [9 \times 2^{2(1+k)}] + 1/8 - 1/2^{3+k}$, $CS(k) = v - 11 / (9 \times 2^{2k+2}) + (1/2^{k-1} - 3) / 8$ and $SW(k) = v - [1 + 1 / (9 \times 2^{2k-1})] / 8$, respectively.

Proof. See Appendix.

When consumers are weakly differentiated, in equilibrium every firm acquires customer flexibility data of any quality and extracts more rents from consumers on its turf. As a result, following data acquisition each firm enjoys higher profits, and the increase in profits is larger, the better is data quality. Both with and without flexibility data every firm follows a monopolization strategy on its turf and serves all consumers there, such that social welfare is always maximized. Higher profits following data acquisition then go hand in hand with a reduction in consumer surplus, and data of a better quality harms consumers more. Figures 6a and 6b present consumer surplus and social welfare in equilibrium as a function of data quality in the two versions of our model.

When consumers are strongly differentiated, in equilibrium only one of the firms acquires customer data if $k = 1$, and both firms acquire data if $k \geq 2$. In the former case the profits of firm A (acquiring customer data) naturally increase. Firm B is worse-off because it loses market shares and charges lower prices on both turfs. The share of consumers buying from the preferred firm gets larger because firm A gains more consumers on the own than on the rival's turf, such that social welfare increases. Consumers are better-off because both total transport costs and their payments to the firms decrease.

If data quality is relatively high, $k \geq 2$, firms end up in the prisoner's dilemma and are worse-off with data acquisition because intensified competition does not allow them to make use of customer data. However, firms' profits decrease less if they acquire data of a better quality. Data acquisition is beneficial for social welfare as the distribution of consumers between the firms becomes more symmetric. If firms acquire perfect customer data, then social welfare is maximized. Consumers benefit from data acquisition, while the benefit is smaller the better is data quality.

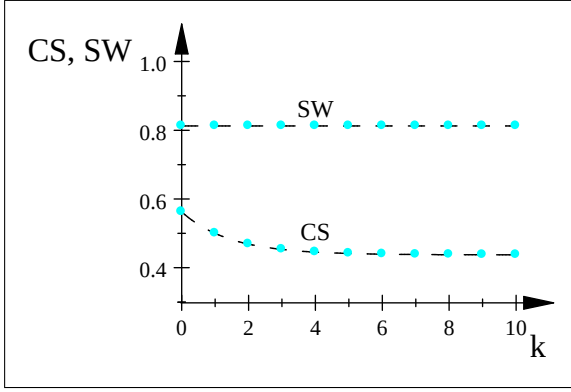


Figure 6a: Weakly differentiated consumers.

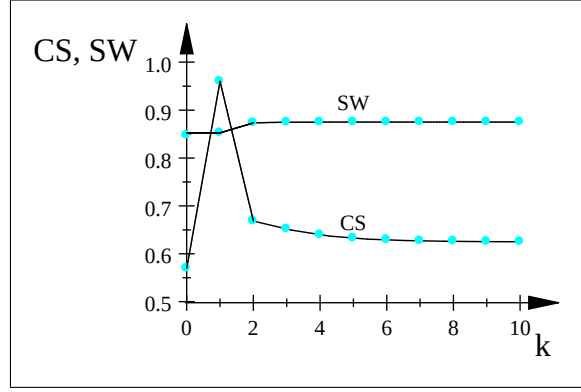


Figure 6b: Strongly differentiated consumers.

Figures 6a-6b: Consumer surplus and social welfare in equilibrium in the two versions of our model. We used the values $v = 1$ and $\underline{t} = 1/2$.

4 Robustness Check and Policy Implications

In the main analysis we considered two extreme cases with respect to consumer heterogeneity in flexibility: *i*) weakly differentiated consumers with $\underline{t} > 0$ and $l \leq 2$ and *ii*) strongly differentiated consumers with $\underline{t} = 0$ and $\lim_{\underline{t} \rightarrow 0} l = \infty$. Here we will consider the remaining case of *moderately differentiated* consumers with $\underline{t} > 0$ and $l > 2$. We will show that in that case depending on consumer heterogeneity firms' incentives to acquire flexibility data in equilibrium are same as in the version of our model with either weakly or strongly differentiated consumers. We will also show that data acquisition may have a different effect on firms' profits, consumer surplus and social welfare in equilibrium compared to the two versions of our model. In the following lemma we characterize the types of consumer flexibility segments depending on data quality when consumers are moderately differentiated.

Lemma 3. (Segment types with moderately differentiated consumers.) *Assume that consumers are moderately differentiated. On any segment $m \geq 2$ consumers are weakly differentiated. On segment $m = 1$ consumers are weakly differentiated if $k \geq \log_2(l - 1)$ and are moderately differentiated otherwise.*

Proof. See Appendix.

Since with the improvement in data quality every flexibility segment is divided into two

segments of equal size, consumers on the upper flexibility segment can be only weakly differentiated. If customer data is of good quality, consumers on segment $m = 1$ are also weakly differentiated and are moderately differentiated otherwise. We showed in Lemma 2 that on its turf on a segment with weakly differentiated consumers a firm follows a monopolization strategy and serves in equilibrium all consumers there. In Lemma 4 we will show that on its turf on a segment with moderately differentiated consumers a firm follows a market-sharing strategy as in the case with strongly differentiated consumers and loses in equilibrium the more flexible consumers to the rival.

Lemma 4. (Equilibrium on a segment with moderately differentiated consumers.)

Assume that both firms hold customer flexibility data of quality $k \geq 0$. Consider some address x on firm i 's turf and some segment m with moderately differentiated consumers. In equilibrium firms charge there prices $p_{im}^(x, k) = [2\bar{t}^m(k) - \underline{t}^m(k)] |1 - 2x| / 3$ and $p_{jm}^*(x, k) = [\bar{t}^m(k) - 2\underline{t}^m(k)] |1 - 2x| / 3$, where firm i serves consumers with $t \geq [\bar{t}^m(k) + \underline{t}^m(k)] / 3$. Firms realize there profits $\Pi_{im}^{D,D}(x, k) = [2\bar{t}^m(k) - \underline{t}^m(k)]^2 |1 - 2x| / [9(1 - \underline{t})]$ and $\Pi_{jm}^{D,D}(x, k) = [\bar{t}^m(k) - 2\underline{t}^m(k)]^2 |1 - 2x| / [9(1 - \underline{t})]$.*

Proof. See Appendix.

Lemma 4 generalizes the results of our main analysis and shows that on some segment on its turf a firm follows a market-sharing strategy not only when consumers are strongly differentiated there, but whenever they are not weakly differentiated. Given the uniform distribution of consumer flexibility, if the transport cost parameter of the most flexible consumer on some segment on a firm's turf is more than two times smaller than the transport cost parameter of the least flexible consumer, then it may be costly for a firm to monopolize that segment. Precisely, if the rival's price is relatively low, a firm optimally targets the less flexible consumers on that segment and loses the more flexible ones to the rival. In the next lemma we characterize the equilibrium in the symmetric subgame with moderately differentiated consumers depending on data quality based on the results of Lemmas 2-4.

Lemma 5. (Symmetric subgame with moderately differentiated consumers.) *Assume that both firms hold customer flexibility data of quality k and consumers are weakly differentiated. The equilibrium depends on the quality of customer data.*

i) If $k < \log_2(l - 1)$, then on x on the turf of firm $i = \{A, B\}$ on $m = 1$ firms charge prices $p_{i1}^(x, k) = \underline{t} [1 + 2(l - 1) / 2^k] |1 - 2x| / 3$ and $p_{j1}^*(x, k) = \underline{t} [(l - 1) / 2^k - 1] |1 - 2x| / 3$,*

where firm i serves consumers with $t \geq \underline{t} [2 + (l - 1) / 2^k] / 3$. On $m \geq 2$ firms charge prices $p_{im}^*(x, k) = \underline{t}^m(k) |1 - 2x|$ and $p_{jm}^*(x, k) = 0$, where firm i serves all consumers. The profit of firm i is

$$\Pi_i^{D,D}(k) = \Pi_i^1(k) := \frac{\underline{t} (2^k - 1) (l + 1)}{2^{3+k}} + \frac{\underline{t}}{36(l - 1)} \left[1 + \frac{2(l - 1)}{2^k} \right]^2 + \frac{\underline{t}}{36(l - 1)} \left(\frac{l - 1}{2^k} - 1 \right)^2$$

ii) If $k \geq \log_2(l - 1)$, then on x on the turf of firm $i = \{A, B\}$ on any m firms charge prices $p_{im}^*(x, k) = \underline{t}^m(k) |1 - 2x|$ and $p_{jm}^*(x, k) = 0$. Every firm serves all consumers on its turf and realizes the profit $\Pi_i^{D,D}(k) = \Pi_i^2(k) : = \underline{t} (l + 1) / 8 - \underline{t} (l - 1) / 2^{3+k}$.

Proof. See Appendix.

Lemma 5 shows that if data quality is low ($k < \log_2(l - 1)$), we get a similar equilibrium as in the case with strongly differentiated consumers. Precisely, every firm loses consumers on its turf on segment $m = 1$ with moderately differentiated consumers, where a firm follows a market-sharing strategy. If $k = \log_2(l - 1)$, consumers on segment $m = 1$ become weakly differentiated and every firm follows a monopolization strategy on any segment on its turf. As a result, for any $k \geq \log_2(l - 1)$ every firm serves all consumers on its turf in equilibrium as in the case with weakly differentiated consumers.

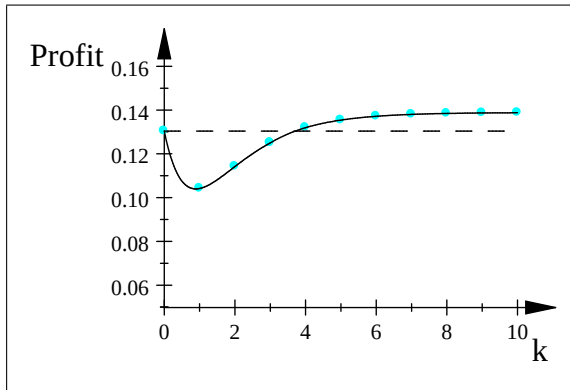


Figure 7a: Profit, $l = 9$.

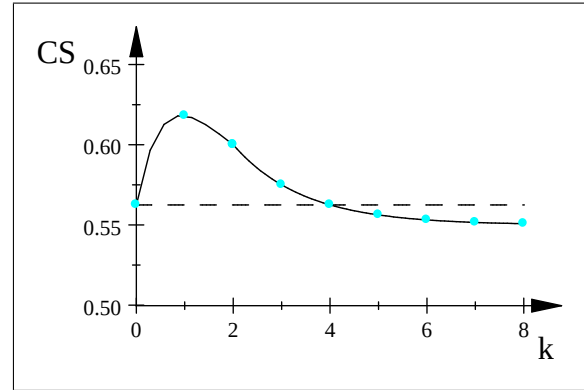


Figure 7b: CS, $l = 5$ and $v = 1$.

Figures 7a-7b: Profit and consumer surplus with moderately differentiated consumers in the symmetric subgame. New results compared to the previous analysis.

How do profits change with the improvement in data quality? If consumers are relatively homogeneous ($l \leq 25/11$), profits increase monotonically in k , similar to the case of weakly

differentiated consumers. Otherwise ($l > 25/11$), profits exhibit a U -shaped relationship in data quality as in the case with strongly differentiated consumers: They decrease when data quality is low ($k \leq \log_2[20(l-1)/(9l+5)]$) and increase afterwards. However, different from that case the negative competition effect does not always dominate with moderately differentiated consumers. If l is relatively small ($l < 8 + 3\sqrt{5}$), we get that $\Pi_i^1(0) < \lim_{k \rightarrow \infty} \Pi_i^2(k)$, such that the positive rent-extraction effect dominates when data quality is good enough (see Figure 7a). This new result, however, supports the conclusion of the main analysis that the profit effect of data acquisition is more likely to be positive when consumers are relatively homogeneous in flexibility. We now turn to the asymmetric subgame, where only firm A holds customer flexibility data. We will describe separately the equilibrium on each turf. The following lemma summarizes our results for the turf of firm A .

Lemma 6. (Asymmetric subgame with moderately differentiated consumers. Firm A 's turf.) *Assume that only firm A holds customer flexibility data of quality $k \geq 1$ and consumers are moderately differentiated. The equilibrium on the turf of firm A depends on consumer heterogeneity in flexibility and the quality of customer data.*

1. If $l > 3$, two cases are possible depending on k .
 1. i) If $k < \log_2(l-1)$, firm B charges $p_B^*(x, k) = \underline{t}(1-2x)[(l-1)/2^{k+1} - 1/2]$ and serves consumers with $t < \underline{t}[3/4 + (l-1)/2^{k+2}]$ on $m = 1$, where firm A charges $p_{A1}^*(x, k) = \underline{t}[1/2 + 3(l-1)/2^{k+1}](1-2x)/2$. On $m \geq 2$ firm A serves all consumers and charges prices $p_{Am}^*(x, k) = \underline{t}[1/2 + (l-1)(2m-1)/2^{k+1}](1-2x)$. Firms realize profits

$$\begin{aligned} \Pi_A(x < 1/2, k) &= \Pi_A^1(x < 1/2, k) := \frac{\underline{t}}{16(l-1)} \left[\frac{1}{2} + \frac{3(l-1)}{2^{k+1}} \right]^2 + \frac{(2^k - 1)(1 + 2^k - \underline{t})}{2^{2k+3}}, \\ \Pi_B(x < 1/2, k) &= \Pi_B^1(x < 1/2, k) := \frac{\underline{t}}{8(l-1)} \left(\frac{l-1}{2^{k+1}} - \frac{1}{2} \right)^2. \end{aligned}$$

1. ii) If $k \geq \log_2(l-1)$, firm B charges $p_B^*(x, k) = 0$ and serves no consumers on firm A 's turf. Firm A charges $p_{Am}^*(x, k) = \underline{t}^m(k)(1-2x)$. Firms realize profits $\Pi_A(x < 1/2, k) = \Pi_A^2(x < 1/2, k) := \underline{t}(1+l)/8 - \underline{t}(l-1)/2^{3+k}$ and $\Pi_B(x < 1/2, k) = \Pi_B^2(x < 1/2, k) := 0$.
2. If $l \leq 3$, for any $k \geq 1$ only case 1. ii) applies.

Proof. See Appendix.

As Lemma 6 shows, the equilibrium depends on l and k , because both determine consumer heterogeneity on a given segment. If l is low ($l \leq 3$), consumers are weakly differentiated on

any flexibility segment (for any $k \geq 1$), such that firm A serves all consumers on its turf in equilibrium. If l is large ($l > 3$), consumers on segment $m = 1$ remain moderately differentiated if data quality is low ($k < \log_2(l - 1)$) and are weakly differentiated otherwise. In the former case firm A loses the more flexible consumers on its turf and serves all consumers there in the latter case. If $l > 3$, the profits of firm B decrease monotonically in k when data quality is low ($k \leq \log_2(l - 1)$), because both its prices and market share become smaller. With a further improvement in data quality ($k > \log_2(l - 1)$) firm B realizes zero profits because it does not serve any consumers on firm A 's turf as in the case $l \leq 3$.

If consumers are relatively homogeneous ($l \leq 3$), the profits of firm A increase monotonically in data quality. In that case the negative competition effect is absent because firm A follows a monopolization strategy on any segment on its turf (for any $k \geq 1$). Then with data of a better quality firm A extracts more rents. If $l > 3$, the effect of data quality on the profits of firm A can be either positive or negative. If $k < \log_2(l - 1)$, the profits of firm A on its turf decrease with the improvement in data quality, because firm A anchors its prices to the uniform prices of firm B , which decrease in data quality. At $k = \log_2(l - 1)$ the uniform prices of firm B get the level of zero, such that with a further improvement in data quality the negative competition effect for firm A 's profits is absent. If $k \geq \log_2(l - 1)$, the profits of firm A increase monotonically in data quality: On each segment firm A follows a monopolization strategy and the improvement in data quality gives rise only to a positive rent-extraction effect. This is different in the case of strongly differentiated consumers, where the profits of firm A on its turf always decreases in data quality, because consumers on segment $m = 1$ remain strongly differentiated for any data quality, which gives rise to a negative competition effect. In the following lemma we state the equilibrium on the turf of firm B .¹⁷

Lemma 7. (Asymmetric subgame with moderately differentiated consumers. Firm B 's turf.) *Assume that only firm A holds customer flexibility data of quality $k \geq 1$ and consumers are moderately differentiated. The equilibrium on the turf of firm A depends on consumer heterogeneity in flexibility and the quality of customer data.*

¹⁷Unfortunately, in the asymmetric subgame with moderately differentiated consumers it is not possible to derive the equilibrium distribution of consumers between the firms for a deliberate l in the case $k \geq \log_2[3(l - 1)/(2l - 4)]$, which explains the usage of the sign “ $\approx$ ” for the profit of firm A in the proposition. However, for any l there is some $\hat{m}$, such that $p_B^*(x, k)/(2x - 1) \in [\underline{t}^{\hat{m}}(k), \bar{t}^{\hat{m}}(k)]$. As there is no analytical solution for $\hat{m} \in N$ depending on l , we will assume that $p_B^*(x, k) = \underline{t}^{\hat{m}}(k)(2x - 1)$. Under this assumption the equilibrium profit of firm A on firm B 's turf is a continuous function of data quality.

1. If $l < 5$, two cases are possible depending on k .

- i) If $1 \leq k < \log_2 [3(l-1)/(2l-4)]$, firm B charges $p_B^*(x, k) = \underline{t} [1 + (l-1)/2^k] (2x-1)$ and realizes the profit $\Pi_B^{D,ND}(x > 1/2, k) = \Pi_B^1(x > 1/2, k) := (\underline{t} - \underline{t}/2^{k+1}) [1 + (l-1)/2^k] / 4$. Firm A serves only consumers on $m = 1$ with $t < \underline{t} [1 + (l-1)/2^{k+1}]$, where it charges prices $p_{A1}^*(x, k) = \underline{t}(l-1)(2x-1)/2^{k+1}$. The profit of firm A is $\Pi_A^{D,ND}(x > 1/2, k) = \Pi_A^1(x > 1/2, k) := \underline{t}(l-1)/2^{2k+4}$.
- ii) If $k \geq \log_2 [3(l-1)/(2l-4)]$, then firm B charges $p_B^*(x, k) = \underline{t} [l/2 + (l-1)/2^{k+2}] (2x-1)$ and realizes the profit $\Pi_B^{D,ND}(x > 1/2, k) = \Pi_B^2(x > 1/2, k) := \underline{t} [l/2 + (l-1)/2^{k+2}]^2 / [4(l-1)]$. On $m \leq \hat{m} - 2$ firm A charges prices $p_{Am}^*(x, k) = p_B^*(x, k) - \bar{t}^m(k) (2x-1)$ and serves all consumers there, where $\hat{m} = 5/4 + 2^k(l-2)/[2(l-1)]$. On $\hat{m} - 1$ firm A charges $p_{A,\hat{m}-1}^*(x, k) = \underline{t}(l-1)(2x-1)/2^{k+1}$, where it serves consumers with $t < \underline{t} + (\bar{t} - \underline{t})(2\hat{m}-3)/2^{k+1}$. On $m \geq \hat{m}$ firm A charges zero prices and does not serve any consumers. Firm A realizes the profit

$$\Pi_A^{D,ND}(x > 1/2, k) \approx \Pi_A^2(x > 1/2, k) := \frac{\underline{t}(l-1)}{2^{2k+4}} + \frac{\underline{t}}{2^7} \left[\frac{2(l-2)}{l-1} + \frac{1}{2^k} \right] \left[2(l-2) - \frac{3(l-1)}{2^k} \right].$$

2. If $l \geq 5$, for any $k \geq 1$ only case 1. ii) applies.

Proof. See Appendix.

We know from the main analysis that firm B loses the more flexible consumers on its turf in the asymmetric subgame even in the version with weakly differentiated consumers (if $l > 3/2$). It is then straightforward that firm B does not serve all consumers on its turf when consumers become more differentiated. With the improvement in data quality available to firm A , firm B is forced to charge lower prices in equilibrium. As a result, its profits decrease. The profits of firm A exhibit a U -shaped relationship in data quality. The equilibrium prices of firm B , which decrease in data quality, put a downward pressure on the prices of firm A , such that its profits first decrease in k . However, as firm B targets on its turf the more loyal consumers, its prices approach some positive values when customer data becomes perfect ($\lim_{k \rightarrow \infty} p_B^*(x, k) = 1/[2(2x-1)]$), which limits the strength of the negative competition effect for firm A 's profits. As a result, when data quality becomes good enough ($k = \log_2 [5(l-1)/(2(l-2))]$), the profits of firm A on the rival's turf start to increase as firm A can extract more rents there. In the next lemma we combine the results of Lemmas 6 and 7 and state firms' profits in the asymmetric subgame on both turfs together.

Lemma 8. (Asymmetric subgame with moderately differentiated consumers. Firms' total profits on both turfs.) Assume that only firm A holds customer flexibility data of quality $k \geq 1$ and consumers are moderately differentiated. Total profits of the firms on both turfs depend on consumer heterogeneity in flexibility and the quality of customer data.

1. If $2 < l \leq 3$, total profits of firm $i = \{A, B\}$ on both turfs take the form

$$\Pi_i^{D,ND}(k) = \begin{cases} \Pi_i^2(x < 1/2, k) + \Pi_i^1(x > 1/2, k) & \text{if } 1 \leq k \leq \log_2 \left[\frac{3(l-1)}{2(l-2)} \right] \\ \Pi_i^2(x < 1/2, k) + \Pi_i^2(x > 1/2, k) & \text{if } k > \log_2 \left[\frac{3(l-1)}{2(l-2)} \right]. \end{cases}$$

2. If $3 < l \leq 7/2$, total profits of firm $i = \{A, B\}$ on both turfs take the form

$$\Pi_i^{D,ND}(k) = \begin{cases} \Pi_i^1(x < 1/2, k) + \Pi_i^1(x > 1/2, k) & \text{if } 1 \leq k < \log_2(l-1) \\ \Pi_i^2(x < 1/2, k) + \Pi_i^1(x > 1/2, k) & \text{if } \log_2(l-1) \leq k < \log_2 \left[\frac{3(l-1)}{2(l-2)} \right] \\ \Pi_i^2(x < 1/2, k) + \Pi_i^2(x > 1/2, k) & \text{if } k \geq \log_2 \left[\frac{3(l-1)}{2(l-2)} \right]. \end{cases}$$

3. If $7/2 < l < 5$, total profits of firm $i = \{A, B\}$ on both turfs take the form

$$\Pi_i^{D,ND}(k) = \begin{cases} \Pi_i^1(x < 1/2, k) + \Pi_i^1(x > 1/2, k) & \text{if } 1 \leq k < \log_2 \left[\frac{3(l-1)}{2(l-2)} \right] \\ \Pi_i^1(x < 1/2, k) + \Pi_i^2(x > 1/2, k) & \text{if } \log_2 \left[\frac{3(l-1)}{2(l-2)} \right] \leq k < \log_2(l-1) \\ \Pi_i^2(x < 1/2, k) + \Pi_i^2(x > 1/2, k) & \text{if } k \geq \log_2(l-1). \end{cases}$$

4. If $l \geq 5$, total profits of firm $i = \{A, B\}$ on both turfs take the form

$$\Pi_i^{D,ND}(k) = \begin{cases} \Pi_i^1(x < 1/2, k) + \Pi_i^2(x > 1/2, k) & \text{if } 1 \leq k < \log_2(l-1) \\ \Pi_i^2(x < 1/2, k) + \Pi_i^2(x > 1/2, k) & \text{if } k \geq \log_2(l-1). \end{cases}$$

Proof. See Appendix.

We know from Lemma 6 that on the turf of firm A firm B does not serve any consumers if they are sufficiently homogeneous ($l \leq 3$). Otherwise ($l > 3$), firm B serves some consumers only if data quality available to firm A is not very precise ($k < \log_2(l-1)$). The profits of firm B decrease in k in that case and take the value of zero at $k = \log_2(l-1)$. As follows from Lemma 7, on the turf of firm B its profits decrease in data quality and approach some positive

value when data becomes perfect. As a result, in the asymmetric subgame with moderately differentiated consumers the profits of firm B decrease monotonically in k .

We know from Lemma 6 that with sufficiently homogeneous consumers ($l \leq 3$), the profits of firm A on its turf increase monotonically in k . They exhibit a U -shaped relationship in data quality on the turf of firm B and on firm A 's turf if $l > 3$. Then the behavior of firm A 's total profits as a function of data quality depends on consumer heterogeneity in flexibility. If $l \leq 3$, total profits of firm A increase monotonically in data quality. If $3 < l \leq 7/2$, they decrease till $k = \log_2(l - 1)$, where firm A turns to a monopolization strategy on segment $m = 1$. With a further improvement in data quality only the rent-extraction effect is present and the profits of firm A on the own turf and its total profits increase in k . If $7/2 < l < 5$, total profits of firm A also exhibit a U -shaped relationship in data quality: They decrease in k till some critical value ($k = \log_2(7/2)$ if $9/2 < l < 5$ and $k = \log_2(l - 1)$ otherwise) and increase afterwards. If $l \geq 5$, the profits of firm A decrease till $k = \log_2(7/2)$ and increase afterwards.

We conclude that if $l \leq 3$, the profits of firm A increase monotonically in data quality (for any $k \geq 1$) as in the case with weakly homogeneous consumers. Otherwise ($l > 3$), they exhibit a U -shaped relationship in data quality as in the case with strongly differentiated consumers. In the next proposition we state firms' equilibrium incentives to acquire customer flexibility data.

Proposition 6. (Incentives to acquire customer flexibility data with moderately differentiated consumers.) *Assume that consumers are moderately differentiated in flexibility. Firms' incentives to acquire customer flexibility data depend on its quality and consumer heterogeneity.*

1. *If $l \leq 3$, in the unique Nash equilibrium (in dominant strategies) both firms acquire data of any quality $k \geq 1$.*
2. *If $l > 3$, the equilibrium depends on the quality of customer data: i) if $k = 1$, two Nash equilibria exist where one of the firms acquires customer data, ii) if $k \geq 2$, the unique Nash equilibrium (in dominant strategies) exists, where both firms acquire customer data.*

Proof. See Appendix.

If consumers are relatively homogeneous in flexibility ($l \leq 3$), in equilibrium both firms acquire customer data of any quality $k \geq 1$, as in the version of our model with weakly differentiated consumers. The reason is that in that case for any $k \geq 1$ consumers on any flexibility segment are weakly differentiated, such that data acquisition gives rise only to a positive rent-extraction

effect (both in the symmetric subgame and on firm A 's turf in the asymmetric subgame). Otherwise ($l > 3$), when data quality is low, consumers on segment $m = 1$ are strongly differentiated, such that data acquisition intensifies competition. As a result, both firms acquire flexibility data in equilibrium only if its quality is sufficiently high ($k \geq 2$), as in the case with strongly differentiated consumers.

Proposition 7. (Profit effects of customer data acquisition with moderately differentiated consumers.) *Assume that consumers are moderately differentiated in flexibility. The profit effect of customer data acquisition depends on the type of the equilibrium.*

1. *Consider the equilibrium, where both firms acquire customer flexibility data. Depending on consumer heterogeneity in flexibility we get the following results.*

- i) *If $2 < l \leq 5$, firms are (weakly) better off following the acquisition of data of any quality.*
- ii) *If $5 < l < 8 + 3\sqrt{5}$, firms are worse off following the acquisition of data provided its quality is low, $k < \hat{k}(l)$, where $\hat{k}(l) := \log_2 [-(9l^2 - 18l + 9) / (l^2 - 16l + 19)]$. If data quality is high, $k \geq \hat{k}(l)$, firms are (weakly) better off.*

iii) *If $l \geq 8 + 3\sqrt{5}$, firms are worse off following the acquisition of data of any quality.*

2. *Consider the equilibrium, where only firm A acquires customer data. Firm A always gains from data acquisition. The effect of data acquisition on the profits of firm B depends on consumer heterogeneity in flexibility.*

- i) *If $3 < l \leq (12\sqrt{15} + 101) / 43$, firm B (weakly) gains from data acquisition.*
- ii) *If $l > (12\sqrt{15} + 101) / 43$, firm B loses from data acquisition.*

Proof. See Appendix.

Proposition 7 supports the results of our main analysis and shows that the profit effects of data acquisition identified in the main analysis are not driven by the extreme ratios of consumer heterogeneity considered there and are valid for other ratios too. Precisely, if l is relatively small ($2 < l \leq 5$), we get the same pattern as in the case with weakly differentiated consumers, where firms are better off with customer data of any quality. If l is relatively large ($l \geq 8 + 3\sqrt{5}$), we get the same pattern as in the case with strongly differentiated consumers, where firms are always worse off from data acquisition. However, for moderate values of l ($5 < l < 8 + 3\sqrt{5}$) we get new results compared to the main analysis (see Figure 7a). In that case the profit effect of data acquisition depends on data quality. If data quality is low ($k < \hat{k}(l)$), the negative competition effect dominates and profits decrease following data acquisition. If

data quality is high enough ($k \geq \widehat{k}(l)$), the positive rent-extraction effect dominates and profits increase following data acquisition. In the next proposition we state consumer surplus and social welfare in equilibrium and the welfare effects of customer data acquisition when consumers are moderately differentiated.¹⁸

Proposition 8. (Welfare effects of customer data acquisition with moderately differentiated consumers.) *Assume that consumers are moderately differentiated and both firms hold customer flexibility data of quality k . Depending on data quality consumer surplus and social welfare are as follows.*

i) If $k < \log_2(l - 1)$, then social welfare and consumer surplus are

$$\begin{aligned} SW &= SW^1(k) := v - \frac{t(l^2 - 3)}{8(l - 1)} - \frac{t}{4(l - 1)} \left(\frac{2}{3} + \frac{l - 1}{3 \times 2^k} \right)^2 \text{ and} \\ CS &= CS^1(k) := v - \frac{t[2^{2k}(27l^2 - 29) + 2^k(16l - 18l^2 + 2) - 44l + 22l^2 + 22]}{72 \times 2^{2k}(l - 1)}. \end{aligned}$$

ii) If $k \geq \log_2(l - 1)$, then social welfare and consumer surplus are $SW = SW^2 := v - t(1 + l)/8$ and $CS = CS^2(k) := v - 3t(1 + l)/8 + t(l - 1)/2^{2+k}$.

Following data acquisition social welfare always increases. The effect of data acquisition on consumer surplus depends on consumer heterogeneity in flexibility and the quality of customer data.

i) If $2 < l < (7 + 3\sqrt{3})/2$, then consumers are (weakly) better off if data quality is low, $k \leq \log_2 \left[9(l - 1)^2 / (-2l^2 + 14l - 11) \right]$, and are worse off otherwise.

ii) If $l \geq (7 + 3\sqrt{3})/2$, then consumers are better off independently of data quality.

Proof. See Appendix.

With moderately differentiated consumers the effect of data acquisition on consumer surplus depends on how strongly consumers differ in flexibility. If $l \geq (7 + 3\sqrt{3})/2$, consumers always benefit from data acquisition independently of its quality. We get the same result as in the case of strongly differentiated consumers, where data acquisition intensifies competition and the rent-extraction effect cannot dominate even when firms acquire perfect customer data. If $2 < l < (7 + 3\sqrt{3})/2$, we get a new result, which differs from those in both ver-

¹⁸In the proposition we will state how consumer surplus and social welfare change in the equilibrium where both firms acquire customer data. We will omit the special case of $k = 1$ and $l > 3$, where in equilibrium only firm A acquires customer data.

sions of our model (see Figure 7b). Precisely, the effect of data acquisition depends on its quality. If data quality is low ($k \leq \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$), consumers benefit from the intensified competition when firms acquire flexibility data. If data quality is high ($k > \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$), with customer data firms are able to extract more rents from consumers making them worse off.

Regarding the effect of data acquisition on social welfare, we should conclude that the result that social welfare does not change following data acquisition (which we get in the case of weakly differentiated consumers) is not robust. The general result is that social welfare increases following data acquisition, because the distribution of consumers between the firms becomes more symmetric. Moreover, up to a certain level of data quality ($k < \log_2(l-1)$), social welfare increases more the better is data quality. In that case we have a contradiction between the maximization of consumer surplus and social welfare. With data of a better quality ($k \geq \log_2(l-1)$), consumers on each flexibility segment are weakly homogeneous and each firm serves all consumers on its turf in equilibrium, such that social welfare is maximized.

The analysis in this section shows that many of our results on firms' incentives to acquire customer flexibility data and their welfare implications are robust, which allows us to make some policy conclusions. While those results depend on consumer heterogeneity in flexibility, in reality it can be difficult to measure it and the intensity of price competition can serve as a proxy. Precisely, price competition is intense in the absence of customer data when consumers are relatively homogeneous in flexibility, in which case a firm targets aggressively the loyal consumers of the rival with (individual or group) discounts. Our results show that in that case following data acquisition the rent-extraction effect dominates, which increases firms' profits and harms consumers. In contrast, customer data intensifies competition when it is initially weak, such that firms' profits decrease and consumers are better-off. When competition is moderate, the trade-off between the rent-extraction and competition effects depends on the quality of customer data: The former dominates if data quality is high, in which case firms' profits increase and consumer surplus gets smaller. Overall, our results imply for consumer policy that data acquisition by the firms is likely to be harmful for consumers if price competition is initially intense or if firms acquire very precise data. In that case there is a contradiction between consumer policy and the policy aimed at the maximization of social welfare, because data acquisition improves the ability of each firm to target its loyal consumers and increases social welfare (or leaves it unchanged).

5 Comparison with the Closest Literature

Corts (1998). Corst (1998) analyzes the profit effect of firms' ability to discriminate among two groups of consumers and shows that it depends on the type of firms' best-response functions. Precisely, the necessary condition for price discrimination to have an unambiguous effect on the equilibrium profits is best-response asymmetry. The profit effect is said to be unambiguous if equilibrium prices change in the same direction for both groups of consumers compared to the uniform price, such that either both decrease or increase. Corst shows that best-response symmetry is a sufficient conditions for price discrimination to have an ambiguous effect on firms' profits, such that compared to the uniform price the equilibrium price increases for one consumer group and decreases for the other. According to Corts, if for any price of the rival a firm charges a higher price to one consumer group than to the other, the same relation should hold for the other firm if best-response symmetry applies, while in all other cases best-response asymmetry holds. In the following lemma we state that our model is characterized by best-response asymmetry.

Lemma 9. *Consider some $x < 1/2$ and segment m . Assume that both firms hold flexibility data of quality k , which improves by one step giving rise to two new segments, m^1 and $m^1 + 1$, with $t \in [\underline{t}^{m^1}(k), \bar{t}^{m^1}(k)]$ and $t \in [\underline{t}^{m^1+1}(k), \bar{t}^{m^1+1}(k)]$, respectively, where $m^1 = 2(m - 1) + 1$. Firms' best-response functions on segments m^1 and $m^1 + 1$ are characterized by best-response asymmetry.*

Proof. See Appendix.

When firms hold data on consumer brand preferences, the ability to discriminate among two flexibility segments gives rise to best-response asymmetry. Consider some address on the turf of firm A . Consumers with that address on a flexibility segment with high transport cost parameters constitute the strong market of firm A , because for any price of firm B it is more costly for them to switch compared to consumers on a segment with low transport cost parameters. Then for any price of the rival firm A charges optimally a higher price to the former group than to the latter. In contrast, consumers on a segment with high transport cost parameters constitute a weak market of firm B , which has to charge a lower price to them. Hence, firms approach the two flexibility segments asymmetrically.

According to Corts, best-response asymmetry is a necessary condition for price discrimination to have an unambiguous effect on firms' prices and profits. However, Corts does not specify

when this effect is positive and when it is negative. Our results show that the sign of the price (profit) effects of the ability to discriminate based on consumer flexibility depends on consumer heterogeneity along that dimension. When consumers are weakly differentiated, the price (profit) effect is positive, such that a firm is able to charge (weakly) higher prices to its loyal consumers on both flexibility segments. When consumers are strongly differentiated, the profit effect is negative because a firm has to charge lower prices to its loyal consumers on both flexibility segments.

Our paper contributes to the literature on competitive price discrimination by identifying conditions for the sign of a profit effect in the discrimination problem characterized by best-response asymmetry. The novelty of our analysis is to consider the case where the profit effect is positive.

Liu and Serfes (2004). Our model is closely related to Liu and Serfes (2004, in the following: LS) who apply a spatial model to investigate firms' incentives to acquire data on consumer brand preferences of various quality for third-degree price discrimination. With data of a better quality firms can identify consumers as belonging to finer location segments. According to the definitions introduced in our analysis, in LS consumers are strongly differentiated (in transport costs), because there is a consumer who can switch brands costlessly (consumer with address $x = 1/2$).¹⁹ This explains the similarity between the results of LS and ours in the version with strongly differentiated consumers. For instance, in LS in the symmetric subgame profits also exhibit a U -shaped relationship in data quality. Precisely, at $k = 1$ each firm follows a market-sharing strategy on its turf because consumers are strongly differentiated there. Data improvement by one step leads to a decrease in firms' profits due to a negative competition effect, as we showed in our analysis. At the next step profits start to increase because the negative competition effect is absent on segment $m = 1$ containing half of all consumers on a firm's turf

¹⁹For the definitions introduced in our analysis to be applicable to both the model of LS and ours, we have to state them in a more general way, related to consumer transport costs. Let $d := t(1 - 2x)$. Then in our model, for any address $x < 1/2$ on the turf of firm A consumers on any segment m with $t \in [\underline{t}^m(k); \bar{t}^m(k)]$ can be characterized by $d \in [\underline{d}_m(k); \bar{d}_m(k)]$, where $\underline{d}_m(k) = \underline{t}^m(k)(1 - 2x)$ and $\bar{d}_m(k) = \bar{t}^m(k)(1 - 2x)$. We say that consumers on segment m are weakly differentiated in transport costs if $\bar{d}_m(k)/\underline{d}_m(k) \leq 2$ and are strongly differentiated if $\underline{d}_m(k) = 0$, in which case $\lim_{\underline{d}_m(k) \rightarrow 0} \bar{d}_m(k)/\underline{d}_m(k) = \infty$. (In a similar way, segments on the turf of firm B can be characterized by $\tilde{d} := t(2x - 1)$). Applying those definitions to LS, we get that for any $k \geq 1$ consumers are strongly differentiated in transport costs on segments $m(k) = 2^{k-1}$ and $m(k) = 2^{k-1} + 1$, because these segments contain a consumer who can switch brands costlessly (with address $x = 1/2$). Differently, for any $k \geq 2$ in LS consumers on segments $m \leq 2^{k-1} - 1$ and $m \geq 2^{k-1} + 2$ are weakly differentiated. Our definitions do not apply only to the case $k = 0$ in LS, where firms cannot distinguish between the loyal consumers of each other.

where a firm sticks to a monopolization strategy.

In the asymmetric subgame in LS the profits of firm A (holding customer data) exhibit a U -shaped relationship in data quality, again exactly as in the version of our model with strongly differentiated consumers. However, in LS profits increase above the initial level when data quality becomes perfect, while in our model they never reach that level. This difference is related to the difference in customer data available to the firms in LS and in our model. In our setup each firm has perfect data on consumer addresses, such that in the asymmetric subgame firm B prices very aggressively the loyal consumers of the rival, and its prices to them approach the level of zero when the quality of flexibility data available to firm A becomes perfect. In LS firm B cannot distinguish among its own loyal consumers and those of the rival in the asymmetric subgame. It then charges a price which decreases in data quality, but approaches some positive value when data becomes perfect as firm B aims to extract rents from its loyal consumers, such that the negative competition effect for the profits of firm A is not that strong.

As a result firms' incentives to acquire customer data and their welfare implications are similar in LS and in the version of our model with strongly differentiated consumers. Precisely, both firms acquire customer data in equilibrium only when its quality is sufficiently high resulting in a prisoner's dilemma for the firms and an increase in consumer surplus. However, different from LS, we get two asymmetric equilibria, where only one of the firms acquires customer data, when data quality is low. This is related to the difference in the underlying equilibrium strategies of the firms. In our model a firm always has a unilateral incentive to acquire customer data, while in LS this is the case only when data quality is sufficiently high.

As expected, in the version with weakly differentiated consumers our results are very different from those of LS. In that case both firms acquire data of any quality in equilibrium, because data acquisition gives rise only to a positive rent-extraction effect. The acquisition of customer flexibility data reduces consumer surplus and leaves social welfare unchanged. In summary, compared to LS, we identify more scope for profitable customer data acquisition, which at the same time harms consumers.

6 Discussion

In this section we discuss how our results depend on the two assumptions: costless data acquisition and perfect data on consumer brand preferences.

Zero costs of data acquisition. In our model we assumed that firms can acquire customer data for free. This assumption allowed us to concentrate on the profit effects of data acquisition. However, in reality customer data has its price, which should influence firms' decisions to acquire it. It is plausible to assume that the costs of producing customer data increase with quality and increase more when quality is better implying a monotonically increasing convex cost function. If we assume a competitive market for customer data, then these assumptions are also valid for the price at which customer data is sold at the market. In the case of weakly differentiated consumers the profit of a firm holding customer data both in the symmetric and asymmetric subgames is a concave function of data quality. Then there should exist some critical value of data quality above which acquiring customer data becomes unprofitable. Hence, in reality firms are likely to acquire only quite imperfect customer data. If consumers are strongly differentiated, then the profit (of a firm with customer data) both in the symmetric and asymmetric subgames first decreases and then starts to increase with the improvement in data quality. Then we can get a similar equilibrium as in the case with weakly differentiated consumers where only data of a relatively low quality is acquired. However, it is also possible that no flexibility data is acquired in equilibrium, because when customer data is of low quality profits are also low and data of a high quality is expensive.

We can conclude that if we introduce convex costs of customer data, then only data of a relatively low quality is acquired in equilibrium. If consumers are strongly differentiated, then the unique equilibrium can exist where firms do not acquire customer data of any quality. In that case the acquisition of customer data (when consumers are weakly differentiated) always harms consumers.

Perfect data on consumer brand preferences. In the main analysis we assumed that firms hold perfect data on consumer brand preferences. Here we will make the weakest assumption on data quality, which only allows a firm to distinguish among its own loyal consumers and those of the rival.²⁰ The distinction between a monopolization and a market-sharing equilibrium strategy introduced in the main analysis is crucial under this assumption too. Precisely, in the former case a firm serves in equilibrium all consumers (regardless of their flexibility) with at least some

²⁰Such data is easy to get, for example, based on consumers' past purchases. Fudenberg and Tirole (2000) consider a two-period model, where in the second period firms can observe from which firm a consumer bought in the first period and price discriminate respectively. They show that in the first period those consumers buy from a firm who have a relatively strong preference for its product.

brand preferences (close to a firm's location) on its turf (see Figure 8). In the latter case a firm serves in equilibrium only the more flexible consumers with any brand preferences on its turf. In the next lemma we will show that if consumers are relatively homogeneous in flexibility on a given segment, then a firm follows a monopolization strategy on that segment on its turf and a market-sharing strategy otherwise.

Lemma 10. *Assume that firms can only distinguish between the loyal consumers of each other and hold flexibility data of quality k . Consider the turf of firm A and segment m . The equilibrium depends on consumer heterogeneity in flexibility.*

i) If consumers are relatively homogeneous on m , $l^m(k) \lesssim 6.7$, firm A follows a monopolization strategy there. Firms charge prices $p_{Am}^(x < 1/2, k) = 2 [\bar{t}^m(k) - \underline{t}^m(k)] / [3 \ln(l^m(k))]$ and $p_{Bm}^*(x < 1/2, k) = [\bar{t}^m(k) - \underline{t}^m(k)] / [3 \ln(l^m(k))]$. Firm A serves consumers with $t(x) \geq [\bar{t}^m(k) - \underline{t}^m(k)] / [3(1 - 2x) \ln(l^m(k))]$. Firms realize profits*

$$\Pi_{Am}(x < 1/2, k) = \frac{4f_t [\bar{t}^m(k) - \underline{t}^m(k)]^2}{18 \ln[l^m(k)]} \text{ and } \Pi_{Bm}(x < 1/2, k) = \frac{f_t [\bar{t}^m(k) - \underline{t}^m(k)]^2}{18 \ln[l^m(k)]}.$$

ii) If consumers are relatively differentiated on m , $l^m(k) > 6.7$, firm A follows a market-sharing strategy there. Firms' prices are implicitly given by the FOCs

$$\begin{aligned} \bar{t}^m(k) - p_{Am}^*(\cdot) + p_{Bm}^*(\cdot) + [2p_{Am}^*(\cdot) - p_{Bm}^*(\cdot)] \ln \left[\frac{p_{Am}^*(\cdot) - p_{Bm}^*(\cdot)}{\bar{t}^m(k)} \right] &= 0 \text{ and} \\ p_{Am}^*(\cdot) - p_{Bm}^*(\cdot) - \underline{t}^m(k) - [p_{Am}^*(\cdot) - 2p_{Bm}^*(\cdot)] \ln \left[\frac{p_{Am}^*(\cdot) - p_{Bm}^*(\cdot)}{\bar{t}^m(k)} \right] &= 0. \end{aligned}$$

Proof. See Appendix.

The intuition behind the results in Lemma 10 is same as in the main analysis. When consumers are relatively homogeneous on some segment on a firm A 's turf, in equilibrium it optimally charges a relatively low price there, which allows to serve consumers with any flexibility at locations close to firm A 's own location. With an increase in consumer flexibility the monopolization strategy becomes costly, firm A turns to a market-sharing strategy and serves in equilibrium only the more flexible consumers on any location on its turf. Different from the main analysis, when firm A follows in equilibrium a monopolization strategy on some segment on its turf, it does not serve all consumers there. The reason is that consumers with address

$x = 1/2$ on any segment on firm A 's turf can switch brand costlessly, such that a monopolization strategy would require the price of zero.

In the symmetric subgame we cannot derive other analytical results above those stated in Lemma 10. To calculate firms' equilibrium profits depending on the quality of flexibility data in the case of relatively homogeneous consumers one would have to rely on approximations using, for instance, Taylor series. To get some intuition on the profit effect of data quality, we compared firms' equilibrium profits under $k = 0$ and $k = 1$ in the case of relatively homogeneous consumers. As in the main analysis, we normalized $\bar{t} = 1$ and considered different values for $\underline{t}$: $1/2$, $1/3$, $1/4$, $1/5$ and $1/6$. In all the cases profits are higher at $k = 1$, which supports the results of the main analysis.

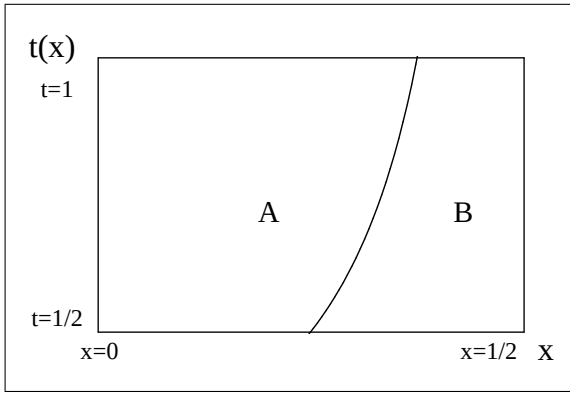


Figure 8a: Firm A follows a monopolization strategy on segment m with $t \in [1/2; 1]$ on its turf.

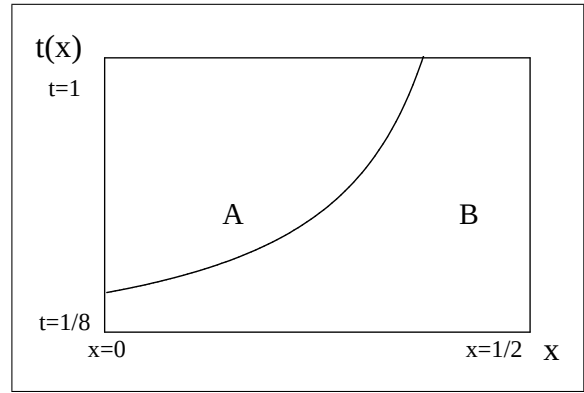


Figure 8b: Firm A follows a market-sharing strategy on segment m with $t \in [1/8; 1]$ on its turf.

Figures 8a-8b: The distribution of consumers between the firms on segment m with $t \in [\underline{t}^m(k); \bar{t}^m(k)]$ on the turf of firm A depending on its equilibrium strategy.

7 Conclusions

In this paper we analyzed the profit effect of price discrimination along consumer flexibility and firms' incentives to acquire data on that characteristic of consumer preferences. We assumed that firms hold perfect data on consumers' brand preferences and can acquire data on their flexibility of an exogenously given quality. In that case price discrimination along consumer flexibility is characterized by best-response asymmetry, such that according to Corts (1998) the

unambiguous profit (price) effect of price discrimination can be either positive or negative. We showed that the sign of the profit effect and firms' incentives to acquire customer data depend on how strongly consumers differ in flexibility. We considered two versions of our model with weakly and strongly differentiated consumers, where in the absence of customer flexibility data a firm follows a monopolization or a market-sharing strategy on its turf, respectively. The type of the equilibrium strategy of a firm on its turf determines in turn the strength of price competition in the absence of customer data. If consumers are weakly differentiated, price competition is initially very strong and cannot be further intensified, such that the acquisition of customer data gives rise only to a positive rent-extraction effect and firms acquire customer data of any quality in equilibrium. If consumers are strongly differentiated, price competition is initially weak and the acquisition of customer data gives rise to both a negative competition effect and a positive rent-extraction effect. As a result, in equilibrium both firms acquire customer data only if its quality is high enough in which case the latter effect is relatively strong. We showed that when consumers are weakly differentiated data acquisition decreases consumer surplus and consumers suffer more when firms acquire data of a better quality. When consumers are strongly differentiated, they are better off following data acquisition and benefit more if firms acquire data of a worse quality.

We also considered the case of moderately differentiated consumers, which proved that our main results are robust. Precisely, firms are likely to benefit from data acquisition if consumers are relatively homogeneous in flexibility or if customer data is of high quality, in which case consumers are worse off. Robust results allow for policy conclusions and the intensity of price competition can serve as a proxy of consumer flexibility. Precisely, if price competition is initially weak data acquisition by the firms is likely to benefit consumers. Finally, we showed that the main intuitions of our analysis apply to the case where firms can only differentiate between two groups of consumers loyal to each of the firms.

We made two main contributions to the literature on competitive price discrimination. First, we analyzed the profit effect of price discrimination along consumer flexibility, which has not yet been intensively covered in the literature. We also derived firms' incentives to acquire customer flexibility data depending on its quality and their welfare effects. Second, we showed that price discrimination in our model is characterized by best-response asymmetry. While the literature has mainly identified a negative profit effect in that case (when firms can only discriminate along

consumer brand preferences), we showed following Corts (1998) that the profit effect can also be positive, provided consumers are relatively homogeneous in flexibility.

8 Appendix

Proof of Lemma 1. *i)* Consider first the case of weakly differentiated consumers. We get that $\text{sign}\{l^m(k) - 2\} = \text{sign}\{(1 - \underline{t})(2 - m) - 2^k \underline{t}\}$. If $m \geq 2$, then $l^m(k) < 2$ for any k . If $m = 1$, then $1 - \underline{t}(1 + 2^k) < 0$ for any $k \geq 0$ as $1 - 2\underline{t} \leq 0$ holds by assumption. Hence, on any segment and for any data quality consumers are weakly differentiated. *ii)* We now turn to the case of strongly differentiated consumers. Consider first $m \geq 2$. For any k it holds $\text{sign}\{l^m(k) - 2\} = \text{sign}\{2 - m\}$. If $m \geq 2$, then $2 - m \leq 0$. On $m = 1$ for any k it holds $\underline{t}^1(k) = 0$. We conclude that for any data quality consumers are strongly differentiated on $m = 1$ and are weakly differentiated on all other segments. *Q.E.D.*

Proof of Lemma 2. As firms are symmetric, we will only consider the turf of firm A . *i)* Consider some $x < 1/2$ and m with weakly differentiated consumers: $t \in [\underline{t}^m(k), \bar{t}^m(k)]$. The best-response function of firm A on x and m is $p_{Am}(x, k, p_{Bm}(x, k)) = p_{Bm}(x, k) + \underline{t}^m(k)(1 - 2x)$, such that firm A serves all consumers for any $p_{Bm}(x, k)$ there. In equilibrium it must hold $p_{Bm}(x, k) = 0$. Indeed, assume that $p_{Bm}(x, k) > 0$. Firm B realizes the profit of zero and has an incentive to decrease its price slightly to gain some consumers. Hence, we get $p_{Am}^*(x, k) = \underline{t}^m(k)(1 - 2x)$ and $p_{Bm}^*(x, k) = 0$. Firm A serves all consumers on m and realizes there a profit $\Pi_{Am}(x, k) = \underline{t}^m(k)(1 - 2x)$. *ii)* Consider now $x > 1/2$ and $m = 1$ with strongly differentiated consumers: $t \in [0, \bar{t}/2^k]$. Firms' best-response functions take the form

$$p_{A1}(x, k, p_{B1}(x, k)) = \begin{cases} p_{B1}(x, k) & \text{if } p_{B1}(x, k) \geq \frac{(1-2x)}{2^k} \\ \frac{p_{B1}(x, k)}{2} + \frac{(1-2x)}{2^{k+1}} & \text{if } p_{B1}(x, k) < \frac{(1-2x)}{2^k} \end{cases}$$

and

$$p_{B1}(x, k, p_{A1}(x, k)) = \begin{cases} p_{A1}(x, k) - \frac{(1-2x)}{2^k} & \text{if } p_{A1}(x, k) \geq \frac{2(1-2x)}{2^k} \\ \frac{p_{A1}(x, k)}{2} & \text{if } p_{A1}(x, k) < \frac{2(1-2x)}{2^k}, \end{cases}$$

which yield equilibrium prices $p_{A1}^*(x, k) = 2(1 - 2x) / (3 \times 2^k)$ and $p_{B1}^*(x, k) = (1 - 2x) / (3 \times 2^k)$. Firm A serves consumers with $t \geq 1 / (3 \times 2^k)$ and $\Pi_{Am}(x, k) = 4(1 - 2x) / (3 \times 2^k)^2$. The profit of firm B is $\Pi_{Bm}(x, k) = (1 - 2x) / (3 \times 2^k)^2$. *Q.E.D.*

Proof of Proposition 1. As firms are symmetric, we will only consider the turf of firm A . *i)* We start with the version of our model with weakly differentiated consumers. From Lemma 1 we know that consumers on all segments are weakly differentiated. Using the results of Lemma

2 we can compute the profits of firm A :

$$\Pi_A^{D,D}(k) = \frac{1}{2^k} \sum_{m=1}^{2^k} \left(\underline{t} + \frac{(1-\underline{t})(m-1)}{2^k} \right) \int_0^{1/2} (1-2x) dx = \frac{1+\underline{t}}{8} - \frac{1-\underline{t}}{2^{3+k}}.$$

Note that these profits hold for any $k \geq 0$. *ii)* We now turn to the version of our model with strongly differentiated consumers. From Lemma 1 we know that on $m = 1$ consumers are strongly differentiated and are weakly differentiated on all other segments. Using the results of Lemma 2 we can compute the profits of firm A summing up both firm's profits on the turf of firm A :

$$\frac{\Pi_A^{D,D}(k)}{\int_0^{1/2} (1-2x) dx} = \frac{4}{9 \times 2^{2k}} + \frac{1}{9 \times 2^{2k}} + \left(\frac{1}{2^k} \right)^2 \sum_{m=2}^{2^k} (m-1) \text{ and } \Pi_A^{D,D}(k) = \frac{10 - 9 \times 2^k}{9 \times 2^{3+2k}} + \frac{1}{8}.$$

Note that these profits hold for any $k \geq 0$. *Q.E.D.*

Proof of Proposition 2. Consider first the turf of firm A and some $x < 1/2$. Given $p_B(x, k)$ firm A maximizes its profit on every segment separately. The best-response function of firm A on segment m is $p_{Am}(x, k, p_B(x, k)) = p_B(x, k) + \underline{t}^m(k)(1-2x)$. Firm B cannot do better than charging $p_B^*(x, k) = 0$, which yields $p_{Am}^*(x, k) = \underline{t}^m(k)(1-2x)$. On its turf firm A realizes the profit

$$\Pi_A(x < 1/2, k) = \frac{1}{2^k} \sum_{m=1}^{2^k} \left(\underline{t} + \frac{(1-\underline{t})(m-1)}{2^k} \right) \int_0^{1/2} (1-2x) dx = \frac{1+\underline{t}}{8} - \frac{1-\underline{t}}{2^{3+k}}, \quad (4)$$

and the profit of firm B on A 's turf is zero.

Consider now the turf of firm B and some $x > 1/2$. Given $p_B(x, k)$ firm A maximizes its profit on every segment separately. The best-response function of firm A on segment m is

$$p_{Am}(x, k, p_B(x, k)) = \begin{cases} 0 & \text{if } p_B(x, k) \leq e \\ \frac{p_B(x, k) - \underline{t}^m(k)(2x-1)}{2} & \text{if } e < p_B(x, k) < f \\ p_B(x, k) - \bar{t}^m(k)(2x-1) & \text{if } p_B(x, k) \geq f, \end{cases}$$

where $e = \underline{t}^m(k)(2x-1)$ and $f = [2\bar{t}^m(k) - \underline{t}^m(k)](2x-1)$. We have to find now the optimal price of firm B given firm A 's best-response function. Note that for any $m \geq 2$ and $k \geq 1$ it holds that $2\bar{t}^{m-1}(k) - \underline{t}^{m-1}(k) = \bar{t}^m(k)$. Assume further that $k \geq 1$. It is straightforward to show that $\Pi_B(x, k, p_B(x, k))$ decreases on $b \leq p_B(x, k) \leq d$. Depending on l and k three

possibilities emerge. First, if $l < 3/2$, then $\Pi_B(x, k, p_B(x, k))$ increases on $p_B(x, k) < a$ and decreases on $a \leq p_B(x, k) < d$, such that $p_B^*(x, k) = \underline{t}(2x - 1)$. For any $x > 1/2$ firm B serves all consumers. Second, if $l > 3/2$ and $k < \log_2 [(l - 1)/(l - 3/2)]$, then $\Pi_B(x, k, p_B(x, k))$ increases on $p_B(x, k) < \underline{t}(2l - 1)(2x - 1)/2$ and decreases on $\underline{t}(2l - 1)(2x - 1)/2 \leq p_B(x, k) < d$, such that $p_B^*(x, k) = \underline{t}(2l - 1)(2x - 1)/2$. For any $x > 1/2$ firm B serves consumers with $t \geq \underline{t}(2l + 1)/4$. Third, if $l > 3/2$ and $k \geq \log_2 [(l - 1)/(l - 3/2)]$, then $\Pi_B(x, k, p_B(x, k))$ increases on $p_B(x, k) < b$ and decreases on $b \leq p_B(x, k) \leq d$, such that $p_B^*(x, k) = \underline{t} [1 + (l - 1)/2^k] (2x - 1)$. For any $x > 1/2$ firm B serves consumers with $t \leq \underline{t} [1 + (l - 1)/2^{k+1}]$.

We now compute firms' profits on the turf of firm B . Consider first $l \leq 3/2$. Firm A serves no consumers on the turf of firm B , the profit of firm B is computed as $\Pi_B^{D,ND}(k) = \int_{1/2}^1 \int_{\underline{t}}^1 [f_{t,x} \underline{t}(2x - 1)] dt dx = \underline{t}/4$. Consider now $l > 3/2$. Assume that $k < \log_2 [(l - 1)/(l - 3/2)]$. The profit of firm B is computed as

$$\Pi_B^{D,ND}(k) = \int_{\frac{1}{2}}^1 (2x - 1) dx \int_{\frac{2+\underline{t}}{4}}^1 \left[f_t \frac{(2 - \underline{t})}{2} \right] dt = \frac{(2l - 1)^2 \underline{t}}{32(l - 1)}.$$

The profit of firm A is computed as

$$\Pi_A(x > 1/2, k) = \int_{\frac{1}{2}}^1 (2x - 1) dx \int_{\underline{t}}^{\frac{2+\underline{t}}{4}} \left[f_t \frac{(2 - 3\underline{t})}{4} \right] dt = \frac{(2l - 3)^2 \underline{t}}{64(l - 1)}. \quad (5)$$

Summing up (4) and (5) we get $\Pi_A^{D,ND}(k) = (12l^2 - 12l + 1) \underline{t} / [64(l - 1)] - (l - 1) \underline{t} / 2^{k+3}$.

Assume finally that $k > \log_2 [(l - 1)/(l - 3/2)]$. The profit of firm B is computed as

$$\Pi_B^{D,ND}(k) = \int_{\frac{1}{2}}^1 (2x - 1) dx \int_{\underline{t} + \frac{1-\underline{t}}{2^{k+1}}}^1 \left[f_t \left(\underline{t} + \frac{1 - \underline{t}}{2^k} \right) \right] dt = \left(1 + \frac{l - 1}{2^k} \right) \left(1 - \frac{1}{2^{k+1}} \right) \frac{\underline{t}}{4}.$$

The profit of firm A is computed as

$$\Pi_A(x > 1/2, k) = \int_{\frac{1}{2}}^1 (2x - 1) dx \int_{\underline{t}}^{\underline{t} + \frac{1-\underline{t}}{2^{k+1}}} \left[f_t \frac{(1 - \underline{t})}{2^{k+1}} \right] dt = \frac{(l - 1) \underline{t}}{2^{2k+4}}. \quad (6)$$

Summing up (4) and (6) we get $\Pi_A^{D,ND}(k) = \underline{t} [(1 + l) / 8 - (l - 1) (2^{k+1} - 1) / 2^{2k+4}]$. We finally note the above derived equilibrium for $k \geq 1$ describes also the equilibrium at $k = 0$. *Q.E.D.*

Proof of Proposition 3. Consider first the turf of firm A . Given $p_B(x, k)$ the best-response

function of firm A on $m = 1$ is

$$p_{A1}(x, k, p_B(x, k)) = \begin{cases} p_B(x, k) & \text{if } p_B(x, k) \geq \frac{1-2x}{2^k} \\ \frac{p_B(x, k)}{2} + \frac{1-2x}{2^{k+1}} & \text{if } p_B(x, k) < \frac{1-2x}{2^k}. \end{cases}$$

The best-response function of firm A on $m \geq 2$ is $p_{Am}(x, k, p_B(x, k)) = p_B(x, k) + \underline{t}^m(k)(1 - 2x)$, such that firm A serves all consumers there. Assume that $p_B(x, k) < (1 - 2x)/2^k$, in which case firm B serves consumers with $t \leq 1/2^{k+1} - p_B(x, k)/[2(1 - 2x)]$ on $m = 1$. Maximization of firm B 's expected profit yields $p_B^*(x, k) = (1 - 2x)/2^{k+1}$ and $p_{A1}^*(x, k) = 3(1 - 2x)/2^{k+2}$. Firm A serves consumers with $t \geq 1/2^{k+2}$. The profit of firm B on the turf of firm A is

$$\Pi_B(x < 1/2, k) = \int_0^{1/2} (1 - 2x) dx \left[\frac{1}{2^{k+2}} \times \frac{1}{2^{k+1}} \right] = \frac{1}{2^{2k+5}}. \quad (7)$$

The profit of firm A on its own turf is

$$\Pi_A(x < 1/2, k) = \left[\frac{2^k - 1}{2^{2k+1}} + \frac{\sum_{n=1}^{2^k-1} n}{2^{2k}} + \frac{9}{2^{2k+4}} \right] \int_0^{1/2} (1 - 2x) dx = \frac{2^k - 1}{2^{k+3}} + \frac{2^{k+3} + 1}{2^{2k+6}}. \quad (8)$$

Consider now the turf of firm B and some $x > 1/2$. On segment m the best-response function of firm A takes the form:

$$p_{Am}(x, k, p_B(x, k)) = \begin{cases} 0 & \text{if } \frac{p_B(x, k)}{(2x-1)} \leq \underline{t}^m(k) \\ \frac{p_B(x, k) - \underline{t}^m(k)(2x-1)}{2} & \text{if } \underline{t}^m(k) < \frac{p_B(x, k)}{(2x-1)} < 2\bar{t}^m(k) - \underline{t}^m(k) \\ p_B(x, k) - \bar{t}^m(k)(2x-1) & \text{if } \frac{p_B(x, k)}{(2x-1)} \geq 2\bar{t}^m(k) - \underline{t}^m(k). \end{cases}$$

The profit of firm B on some $x > 1/2$ is

$$\Pi_B(x, k, p_B(x, k)) = \begin{cases} \left[1 - \frac{p_B(x, k)}{2(2x-1)} \right] p_B(x, k) & \text{if } 0 \leq p_B(x, k) < a \\ \left[1 + \frac{1}{2^{k+1}} - \frac{p_B(x, k)}{2x-1} \right] p_B(x, k) & \text{if } a \leq p_B(x, k) < b \\ \left[\frac{1}{2} + \frac{1}{2^{k+1}} - \frac{p_B(x, k)}{2(2x-1)} \right] p_B(x, k) & \text{if } b \leq p_B(x, k) < c \\ 0 & \text{if } p_B(x, k) \geq c, \end{cases}$$

where $a = (2x - 1)/2^k$, $b = (2x - 1)$, $c = (1 + 1/2^k)(2x - 1)$. Note that $\Pi_B(x, k, p_B(x, k))$ increases on $0 \leq p_B(x, k) < a$, decreases on $b \leq p_B(x, k) < c$ and gets its maximum at $p_B^*(x, k) = (1/2 + 1/2^{k+2})(2x - 1)$ on $a \leq p_B(x, k) < b$. For any $k \geq 1$ it holds $p_B^*(x, k)/(2x - 1) \in$

$[1/2, 1/2 + 1/2^k]$. Assume further that $k = 1$. On $m = 2^{k-1} + 1$ firm B serves consumers with $t \geq 1/2 + 1/2^{k+3}$, on $m = 2^{k-1}$ firm B serves consumers with $t \geq 1/2 - 3/2^{k+3}$. If $k \geq 2$, then firm B serves all consumers on $m \geq 2^{k-1} + 2$ and firm A serves all consumers on $m \leq 2^{k-1} - 1$. The profit of firm B on its turf is

$$\Pi_B(x > 1/2, k) = \int_{1/2}^1 (2x - 1) dx \left(\frac{1}{2} + \frac{1}{2^{k+2}} \right)^2 = \frac{1}{16} \left(1 + \frac{1}{2^k} + \frac{1}{2^{2k+2}} \right). \quad (9)$$

Summing up (7) and (9) we get $\Pi_B^{D,ND}(k) = 1/16 + 1/2^{k+4} + 3/2^{2k+6}$, for any $k \geq 1$.

We now compute the equilibrium prices of firm A on the turf of firm B . We get $p_{Am}^*(x, k) = 5(2x - 1)/2^{k+3}$ if $m = 2^{k-1}$ and $p_{Am}^*(x, k) = (2x - 1)/2^{k+3}$ if $m = 2^{k-1} + 1$. If $k \geq 2$, then on $m \leq 2^{k-1} - 1$ firm A charges $p_{Am}^*(x, k) = p_B^*(x, k) - \bar{t}^m(k)(2x - 1)$. The profit of firm A on the rival's turf is

$$\frac{\Pi_A(x > 1/2, k)}{\int_{1/2}^1 (2x - 1) dx} = \frac{1}{2^{2k+6}} + \frac{25}{2^{2k+6}} + \left(\frac{1}{2} + \frac{1}{2^{k+2}} \right) \left(\frac{2^{k-1} - 1}{2^k} \right) - \frac{\sum_1^{2^{k-1}-1} n}{2^{2k}},$$

such that $\Pi_A(x > 1/2, k) = (2^{2k+2} - 2^{k+2} + 5)/2^{2k+7}$. Summing up $\Pi_A(x > 1/2, k)$ and $\Pi_A(x < 1/2, k)$ in (8) we get $\Pi_A^{D,ND}(k) = (5 \times 2^{2k+2} - 2^{k+2} + 7)/2^{2k+7}$, for any $k \geq 1$.

We now derive the equilibrium on the turf of firm B for $k = 0$. The best-response function of firm A is

$$p_A(x, p_B(x, 0)) = \begin{cases} \frac{p_B(x, 0)}{2} & \text{if } p_B(x, 0) \leq 2(2x - 1) \\ p_B(x, 0) - (2x - 1) & \text{if } p_B(x, 0) > 2(2x - 1). \end{cases}$$

Assume that $p_B(x, 0) \leq 2(2x - 1)$ for any $x > 1/2$, such that firm B serves consumers with $t \geq p_B(x, 0)/[2(2x - 1)]$. Maximization of firm B 's profit yields $p_B^*(x, 0) = (2x - 1) < 2(2x - 1)$ and $p_A^*(x, 0) = (2x - 1)/2$. Firm B serves consumers with $t \geq 1/2$. On the turf of firm B firms realize profits $\Pi_A(x > 1/2, 0) = 1/16$ and $\Pi_B(x > 1/2, 0) = 1/8$. Summing up these profits with (8) and (7) for $k = 0$ we get $\Pi_A^{D,ND}(0) = 13/64$ and $\Pi_B^{D,ND}(0) = 5/32$, respectively. *Q.E.D.*

Proof of Proposition 4. We prove first part *i*) of the proposition. If a firm unilaterally acquires customer data, it keeps all consumers on its turf and extracts more rents from them, it also gains consumers on the rival's turf (if $l > 3/2$), such that the unilateral acquisition of data is always profitable. Assume now that firm A has customer data. We show that firm B always has an incentive to acquire data. When both firms hold data of quality $k \geq 1$, firm B realizes the

profit $\Pi_B^{D,D}(k) = \underline{t} [1/4 + (l-1)(1/8 - 1/2^{3+k})]$, as stated in Proposition 1. Assume first that $l \leq 3/2$, in which case the profit of firm B is $\Pi_B^{D,ND}(k) = \underline{t}/4$ if only firm A holds data, as stated in Proposition 2. The comparison shows $\Pi_B^{D,D}(k) - \Pi_B^{D,ND}(k) = \underline{t}(l-1)(1 - 1/2^k)/8 > 0$, for any $k \geq 1$, such that firm B has an incentive to acquire data when the rival holds it if $l \leq 3/2$. Assume now that $l > 3/2$ and $k < \log_2[(l-1)/(l-3/2)]$, in which case the profit of firm B is $\Pi_B^{D,ND}(k) = \underline{t}(2l-1)^2/[32(l-1)]$ if only the rival holds data, as stated in Proposition 2. The comparison shows

$$\frac{\Pi_B^{D,D}(k) - \Pi_B^{D,ND}(k)}{\underline{t}} = -\frac{(3-2l)^2}{32(l-1)} + \frac{(l-1)}{8} \left(1 - \frac{1}{2^k}\right) \geq -\frac{(2l^2 - 8l + 7)}{32(l-1)}. \quad (10)$$

The inequality in (10) follows from $1/2^k \leq 1/2$ for any $k \geq 1$. Note that $2l^2 - 8l + 7 < 0$ for any $3/2 < l \leq 2$, such that $\Pi_B^{D,D}(k) > \Pi_B^{D,ND}(k)$, and firm B has an incentive to acquire data when the rival holds it if $l > 3/2$ and $k < \log_2[(l-1)/(l-3/2)]$ hold. Assume finally that $l > 3/2$ and $k \geq \log_2[(l-1)/(l-3/2)]$, in which case the profit of firm B is $\Pi_B^{D,ND}(k) = \underline{t} [1 + (l-1)/2^k] (1 - 1/2^{k+1})/4$ if only the rival holds data, as stated in Proposition 2. The comparison shows

$$\Pi_B^{D,D}(k) - \Pi_B^{D,ND}(k) = \frac{\underline{t}}{8} \left[(l-1) \left(1 - \frac{3}{2^k} + \frac{1}{2^{2k}}\right) + \frac{1}{2^k} \right]. \quad (11)$$

The expression in (11), $1 - 3/2^k + 1/2^{2k}$, can be either positive or negative. If $1 - 3/2^k + 1/2^{2k} \geq 0$, then $\Pi_B^{D,D}(k) > \Pi_B^{D,ND}(k)$ for any $3/2 < l \leq 2$. Assume that $1 - 3/2^k + 1/2^{2k} < 0$. Then for any $3/2 < l \leq 2$ it holds that

$$\Pi_B^{D,D}(k) - \Pi_B^{D,ND}(k) \geq \frac{\underline{t}}{8} \left(1 - \frac{2}{2^k} + \frac{1}{2^{2k}}\right), \quad (12)$$

where we derived the RHS of (12) by plugging $l = 2$ into the RHS of (11). The derivative of the RHS of (12) with respect to k is positive for any $k \geq 1$, hence, $\Pi_B^{D,D}(k) - \Pi_B^{D,ND}(k) > \Pi_B^{D,D}(1) - \Pi_B^{D,ND}(1) = \underline{t}/32$. It follows that firm B has an incentive to acquire data when the rival holds it and $l > 3/2$ and $k \geq \log_2[(l-1)/(l-3/2)]$ hold. Hence, firm B has always an incentive to acquire data when the rival holds it. We conclude that there is the unique equilibrium (in dominant strategies) for any $k \geq 1$, where both firms acquire data.

We now prove part *ii*) of the proposition. Assume that firm B does not hold customer data. We analyze the incentives of firm A to acquire data. If firm A does not acquire data,

then $\Pi_A^{D,D}(0) = 5/36$, as stated in Proposition 1. If firm A acquires data, then $\Pi_A^{D,ND}(k) = (5 \times 2^{2k+2} - 2^{k+2} + 7) / 2^{2k+7}$, as stated in Proposition 3. The comparison yields

$$\left[\Pi_A^{D,ND}(k) - \Pi_A^{D,D}(0) \right] \times 9 \times 2^{2k+7} = 5 \times 2^{2k+2} - 9 \times 2^{k+2} + 63. \quad (13)$$

Taking the derivative of the RHS of (13) with respect to k we get $2^{k+2} (\ln 2) (10 \times 2^k - 9) > 0$ for any $k \geq 0$. Hence, for any $k \geq 1$ it holds that $\Pi_A^{D,ND}(k) > \Pi_A^{D,D}(0)$, and a firm always has a unilateral incentive to acquire customer data.

Assume now that firm A holds customer data. We analyze whether firm B also has an incentive to acquire data. If it does not acquire, its profit is $\Pi_B^{D,ND}(k) = (1 + 1/2^k + 3/2^{2k+2}) / 16$, as stated in Proposition 3, while if it acquires data its profit is $\Pi_B^{D,D}(k) = 5 / (9 \times 2^{2(1+k)}) + 1/8 - 1/2^{3+k}$, as stated in Proposition 1. The comparison yields

$$\left[\Pi_B^{D,D}(k) - \Pi_B^{D,ND}(k) \right] \times 9 \times 2^{2k+6} = 36 \times 2^{2k} - 108 \times 2^k + 53. \quad (14)$$

Taking the derivative of the RHS of (14) with respect to k we get $9 \times 2^{k+2} (\ln 2) (2^{k+1} - 3)$, which is negative if $k = 0$ and positive if $k \geq 1$. Evaluating the RHS of (14) at $k = 1$ we get -19 , and at $k = 2$ we get 187 . Hence, if $k = 1$, then a firm does not acquire data when the rival holds it, and acquires it if $k \geq 2$.

We conclude that if $k = 1$, then there are two Nash equilibria where only one of the firms acquires data. If $k \geq 2$, there is a unique Nash equilibrium (in dominant strategies) where both firms acquire data. *Q.E.D.*

Proof of Proposition 5. *i)* For any $k \geq 0$ in equilibrium every firm acquires customer data and serves all consumers on its turf. Then social welfare is computed as $SW^{D,D}(k) = v - 2 \int_0^{1/2} \int_{\underline{t}}^1 [f_t t x] dt dx = v - \underline{t}(l+1)/8$. Consumer surplus is computed by subtracting profits (stated in Proposition 1) from social welfare: $CS^{D,D}(k) = v - 3\underline{t}(l+1)/8 + \underline{t}(l-1)/2^{k+2}$. *ii)* If $k = 0$, then social welfare is computed as

$$SW^{D,D}(0) = v - 2 \int_0^{1/2} \int_{\frac{1}{3}}^1 [f_{t,x} t x] dt dx - 2 \int_0^{1/2} \int_0^{\frac{1}{3}} [f_{t,x} t (1-x)] dt dx = v - 11/72.$$

Consumer surplus can be computed through subtracting profits (stated in Proposition 1) from social welfare: $CS^{D,D}(0) = v - 31/72$. If $k = 1$, then only one firm acquires customer data in

equilibrium, such that $\Pi_A^{D,ND}(1) = 79/512$ and $\Pi_B^{D,ND}(1) = 27/256$. Social welfare is computed as

$$\begin{aligned} SW^{D,ND}(1) &= v - \int_0^{1/21/8} \int_0^1 [f_t t (1-x)] dt dx - \int_0^{1/2} \int_{1/8}^1 [f_t t x] dt dx - \int_{1/2}^1 \int_0^{5/16} [f_t t x] dt dx \\ &\quad - \int_{1/25/16}^1 \int_0^{1/2} [f_t t (1-x)] dt dx - \int_{1/2}^1 \int_{1/2}^{9/16} [f_t t x] dt dx - \int_{1/29/16}^1 \int_{1/2}^{\bar{t}} [f_t t (1-x)] dt dx \\ &= v - 151/1024. \end{aligned}$$

Subtracting profits from social welfare we get $CS^{D,ND}(1) = v - 417/1024$. If $k \geq 2$, then both firms acquire customer data in equilibrium. Social welfare is computed as

$$\begin{aligned} SW^{D,D}(k) &= v - 2 \int_0^{1/2} \int_{\frac{1}{3 \cdot 2^k}}^1 [f_t t x] dt dx - 2 \int_0^{1/2} \int_0^{\frac{1}{3 \cdot 2^k}} [f_t t (1-x)] dt dx \\ &= v - \frac{1}{8} \left(1 + \frac{1}{9 \times 2^{2k-1}} \right). \end{aligned}$$

Subtracting profits (stated in Proposition 1) from social welfare we get consumer surplus

$$CS^{D,D}(k) = v - \frac{11}{9 \cdot 2^{2k+2}} + \frac{1}{8} \left(\frac{1}{2^{k-1}} - 3 \right).$$

Q.E.D.

Proof of Lemma 3. We have to compare the ratio $\bar{t}^m(k)/\underline{t}^m(k)$ with 2. We get that $\text{sign} \{ \bar{t}^m(k)/\underline{t}^m(k) - 2 \} = \text{sign} \{ \underline{t}(l-1)(2-m) - \underline{t} \times 2^k \}$. If $m \geq 2$, then $\underline{t}(l-1)(2-m) - \underline{t} \times 2^k < 0$ for any k , such that consumers are weakly differentiated on segments $m \geq 2$. Consider now $m = 1$. If $k \geq \log_2(l-1)$, then $\underline{t}(l-1) - \underline{t} \times 2^k \leq 0$ and consumers are weakly differentiated on $m = 1$. If $k < \log_2(l-1)$, then $\underline{t}(l-1) - \underline{t} \times 2^k > 0$ and consumers are moderately differentiated on $m = 1$. Finally, note that for any $l > 2$ it holds that $\log_2(l-1) > 0$.

Proof of Lemma 4. As firms are symmetric, we will only consider the turf of firm A . For any $x < 1/2$ on some m firm A serves consumers with $t \geq \tilde{t}_m(x; p_{Am}(x, k), p_{Bm}(x, k))$, where $\tilde{t}_m(x; p_{Am}(x, k), p_{Bm}(x, k)) := [p_{Am}(x, k) - p_{Bm}(x, k)] / (1 - 2x)$ if $\tilde{t}_m(\cdot) \in [\underline{t}^m(k); \bar{t}^m(k)]$. Maximization of firm A 's profit on address x on segment m yields the best-response function

$$p_{Am}(x, k, p_{Bm}(x, k)) = \begin{cases} p_{Bm}(x, k) + \underline{t}^m(k)(1-2x) & \text{if } \frac{p_{Bm}(x, k)}{(1-2x)} \geq \bar{t}^m(k) - 2\underline{t}^m(k) \\ \frac{p_{Bm}(x, k) + \bar{t}^m(k)(1-2x)}{2} & \text{if } \frac{p_{Bm}(x, k)}{(1-2x)} < \bar{t}^m(k) - 2\underline{t}^m(k). \end{cases} \quad (15)$$

Maximization of firm B 's profit on address x on segment m yields the best-response function

$$p_{Bm}(x, k, p_{Am}(x, k)) = \begin{cases} p_{Am}(x, k) - \bar{t}^m(k)(1-2x) & \text{if } \frac{p_{Am}(x, k)}{(1-2x)} \geq 2\bar{t}^m(k) - \underline{t}^m(k) \\ \frac{p_{Am}(x, k) - \underline{t}^m(k)(1-2x)}{2} & \text{if } \underline{t}^m(k) < \frac{p_{Am}(x, k)}{(1-2x)} < 2\bar{t}^m(k) - \underline{t}^m(k) \\ 0 & \text{if } \frac{p_{Am}(x, k)}{(1-2x)} \leq \underline{t}^m(k). \end{cases} \quad (16)$$

Combining (15) and (16) we conclude that two types of equilibria are possible where either firm A monopolizes segment m on x or both firms serve consumers there. In the former equilibrium prices are $p_{Bm}^*(x, k) = 0$ and $p_{Am}^*(x, k) = \underline{t}^m(k)(1-2x)$. For this equilibrium to exist, it must hold that $p_{Bm}^*(x, k) \geq [\bar{t}^m(k) - 2\underline{t}^m(k)](1-2x)$. The latter is, however, wrong because $\bar{t}^m(k)/\underline{t}^m(k) > 2$ holds by assumption. Consider now the latter equilibrium, where $p_{Am}^*(x, k) = [2\bar{t}^m(k) - \underline{t}^m(k)](1-2x)/3$ and $p_{Bm}^*(x, k) = [\bar{t}^m(k) - 2\underline{t}^m(k)](1-2x)/3$. For this equilibrium to exist, it must hold that $p_{Bm}^*(x, k) < [\bar{t}^m(k) - 2\underline{t}^m(k)](1-2x)$ and $\underline{t}^m(k) < p_{Am}^*(x, k)/(1-2x) < 2\bar{t}^m(k) - \underline{t}^m(k)$. Note that given $\bar{t}^m(k)/\underline{t}^m(k) > 2$ both conditions are fulfilled. We conclude that in equilibrium firms share segment m on address x where firm A serves consumers with $t \geq [\bar{t}^m(k) + \underline{t}^m(k)]/3$ and firms realize profits $\Pi_{Am}^{D,D}(x, k) = [2\bar{t}^m(k) - \underline{t}^m(k)]^2(1-2x)/[9(1-\underline{t})]$ and $\Pi_{Bm}^{D,D}(x, k) = [\bar{t}^m(k) - 2\underline{t}^m(k)]^2(1-2x)/[9(1-\underline{t})]$. *Q.E.D.*

Proof of Lemma 5. Since firms are symmetric, we will only consider the turf of firm A . *i)* As we know from Lemma 3, consumers are weakly differentiated on $m \geq 2$ and are moderately differentiated on $m = 1$. We know from Lemma 2 that on any $x < 1/2$ in the former case firms charge prices $p_{Am}^*(x, k) = \underline{t}^m(k)(1-2x)$ and $p_{Bm}^*(x, k) = 0$, and firm A serves all consumers on $m \geq 2$. We know from Lemma 4 that on $m = 1$ firms charge prices $p_{A1}^*(x, k) = [2\bar{t}^1(k) - \underline{t}](1-2x)/3$ and $p_{B1}^*(x, k) = [\bar{t}^1(k) - 2\underline{t}](1-2x)/3$, where firm A serves consumers with $t \geq [\bar{t}^1(k) + \underline{t}]/3$. Since firms are symmetric, the profits of both firms on the turf of firm A are equal to the total profits of firm A on both turfs. Hence,

$$\begin{aligned} & \Pi_A^{D,D}(k) \\ &= \left[\frac{1}{2^k} \sum_2^{2^k} \underline{t}^m(k) + \frac{\underline{t}}{9(l-1)} \left(1 + \frac{2(l-1)}{2^k} \right)^2 + \frac{\underline{t}}{9(l-1)} \left(\frac{l-1}{2^k} - 1 \right)^2 \right] \int_0^{1/2} (1-2x) dx \\ &= \frac{\underline{t}(2^k-1)(l+1)}{2^{k+3}} + \frac{\underline{t}}{36(l-1)} \left[1 + \frac{2(l-1)}{2^k} \right]^2 + \frac{\underline{t}}{36(l-1)} \left(\frac{l-1}{2^k} - 1 \right)^2. \end{aligned}$$

ii) We know from Lemma 3 that consumers are weakly differentiated on any $m \geq 1$. We know from Lemma 2 that on any $x < 1/2$ on any m firms charge prices $p_{Am}^*(x, k) = \underline{t}^m(k)(1 - 2x)$ and $p_{Bm}^*(x, k) = 0$, and firm A serves all consumers on its turf. The profit of firm A is computed as

$$\Pi_A^{D,D}(k) = \frac{1}{2^k} \sum_1^{2^k} \underline{t}^m(k) \int_0^{1/2} (1 - 2x) dx = \frac{\underline{t}(l+1)}{8} - \frac{\underline{t}(l-1)}{2^{3+k}}.$$

Q.E.D.

Proof of Lemma 6. Consider some $x < 1/2$ on the turf of firm A . i) We know from Lemma 3 that if $k < \log_2(l-1)$, then on $m \geq 2$ consumers are weakly differentiated and are moderately differentiated on $m = 1$. Firm B can only gain consumers on $m = 1$, where the best-response function of firm A takes the form

$$p_{A1}(x, k, p_B(x, k)) = \begin{cases} p_B(x, k) + \underline{t}(1 - 2x) & \text{if } p_B(x, k) \geq [\bar{t}^1(k) - 2\underline{t}](1 - 2x) \\ \frac{p_B(x, k) + \bar{t}^1(k)(1 - 2x)}{2} & \text{if } p_B(x, k) < [\bar{t}^1(k) - 2\underline{t}](1 - 2x). \end{cases}$$

Assume that in equilibrium firm B charges $p_B(x, k) < [\bar{t}^1(k) - 2\underline{t}](1 - 2x)$. Then its profit is

$$\Pi_B(x, k) = \left[\frac{\bar{t}^1(k)}{2} - \frac{p_B(x, k)}{2(1 - 2x)} - \underline{t} \right] \frac{p_B(x, k)}{1 - \underline{t}}. \quad (17)$$

Maximizing (17) with respect to $p_B(x, k)$ we get $p_B^*(x, k) = \underline{t}(1 - 2x)[(l-1)/2^{k+1} - 1/2]$ (note that $p_B^*(x, k) < [\bar{t}^1(k) - 2\underline{t}](1 - 2x)$ holds), and the profit of firm B on firm A 's turf is

$$\Pi_B(x < 1/2, k) = \frac{\underline{t}}{8(l-1)} \left(\frac{l-1}{2^{k+1}} - \frac{1}{2} \right)^2.$$

We now compute the profit of firm A . On $m \geq 2$ firm A charges the price

$$p_{Am}^*(x, k) = \underline{t} \left[1/2 + (l-1)(2m-1)/2^{k+1} \right] (1 - 2x)$$

and on $m = 1$ firm A charges $p_{A1}^*(x, k) = \underline{t} [1/2 + 3(l-1)/2^{k+1}] (1 - 2x)/2$, where it serves

consumers with $t \geq \underline{t} [3/4 + (l-1)/2^{k+2}]$. The profit of firm A on its turf is

$$\begin{aligned} \frac{\Pi_A(x < 1/2, k)}{\int_0^{1/2} (1-2x) dx} &= \frac{\underline{t}}{l-1} \left[\frac{1}{4} \left(\frac{1}{2} + \frac{3(l-1)}{2^{k+1}} \right)^2 + \frac{(l-1)}{2^k} \sum_2^{2^k} \left(\frac{1}{2} + \frac{(l-1)(2m-1)}{2^{k+1}} \right) \right] \text{ and} \\ \Pi_A(x < 1/2, k) &= \frac{\underline{t}}{16(l-1)} \left(\frac{1}{2} + \frac{3(l-1)}{2^{k+1}} \right)^2 + \frac{\underline{t}(2^k-1)[l(1+2^k)-1]}{2^{2k+3}}. \end{aligned}$$

ii) We know from Lemma 3 that if $k \geq \log_2(l-1)$, then consumers on all segments are weakly differentiated, in which case firm A monopolizes any segment. Firm B cannot do better than charging $p_B^*(x, k) = 0$, and the price of firm A on m is $p_{Am}^*(x, k) = \underline{t}^m(k)(1-2x)$. The profit of firm A on $x < 1/2$ is

$$\Pi_A(x, k) = \frac{(1-2x)}{2^k} \sum_1^{2^k} \underline{t}^m(k) = \underline{t} \left(\frac{1+l}{2} - \frac{l-1}{2^{k+1}} \right) (1-2x).$$

Total profits of firm A on its turf are $\Pi_A(x < 1/2, k) = \underline{t} [(1+l)/8 - (l-1)/2^{k+3}]$, while firm B realizes the profit of zero.

Note finally that $\log_2(l-1) \leq 1$ if $l \leq 3$ (with the opposite inequality otherwise). *Q.E.D.*

Proof of Lemma 7. Consider some $x > 1/2$ on the turf of firm B . We first derive the best-response function of firm A on some m . On any m for any $x > 1/2$ firm A serves consumers with $t < \tilde{t}_m(x; p_B(x, k), p_{Am}(x, k)) := [p_B(x, k) - p_{Am}(x, k)] / (2x-1)$, provided $\tilde{t}_m(\cdot) \in [\underline{t}^m(k); \bar{t}^m(k)]$. Maximization of firm A 's profit on segment m yields the best-response function

$$p_{Am}(x, k, p_B(x, k)) = \begin{cases} p_B(x, k) - \bar{t}^m(k)(2x-1) & \text{if } \frac{p_B(x, k)}{(2x-1)} \geq 2\bar{t}^m(k) - \underline{t}^m(k) \\ \frac{p_B(x, k) - \underline{t}^m(k)(2x-1)}{2} & \text{if } \underline{t}^m(k) < \frac{p_B(x, k)}{(2x-1)} < 2\bar{t}^m(k) - \underline{t}^m(k) \\ 0 & \text{if } \frac{p_B(x, k)}{(2x-1)} \leq \underline{t}^m(k). \end{cases} \quad (18)$$

Note that for any $m \geq 2$ and $k \geq 1$ it holds that $2\bar{t}^{m-1}(k) - \underline{t}^{m-1}(k) = \bar{t}^m(k)$. Given (18), the

profit function of firm B for some $x > 1/2$ takes the form

$$\Pi_B(x, k, p_B(x, k)) = \begin{cases} p_B(x, k) & \text{if } p_B(x, k) < a \\ \left[\frac{2l-1}{2} - \frac{p_B(x, k)}{2(2x-1)} \right] \frac{p_B(x, k)}{l-1} & \text{if } a \leq p_B(x, k) < b \\ \left[l + \frac{l-1}{2^{k+1}} - \frac{p_B(x, k)}{2x-1} \right] \frac{p_B(x, k)}{l-1} & \text{if } b \leq p_B(x, k) < c \\ \left[\frac{l}{2} + \frac{l-1}{2^{k+1}} - \frac{p_B(x, k)}{2(2x-1)} \right] \frac{p_B(x, k)}{l-1} & \text{if } c \leq p_B(x, k) < d \\ 0 & \text{if } p_B(x, k) \geq d, \end{cases}$$

where $a = \underline{t}(2x-1)$, $b = \underline{t} [1 + (l-1)/2^k] (2x-1)$, $c = (2x-1)$ and $d = \underline{t} [l + (l-1)/2^k] (2x-1)$. It is straightforward to show that for any $k \geq 0$ and any l , $\Pi_B(\cdot)$ decreases on $c \leq p_B(\cdot) < d$. Note next that for any l it holds that $0 < \log_2 [(l-1)/(l-3/2)] < \log_2 [3(l-1)/(2l-4)]$. The profit-maximizing price depends on k . We have to distinguish between three cases. *i)* If $k \leq \log_2 [(l-1)/(l-3/2)]$, then $\Pi_B(\cdot)$ decreases on $\underline{t}(2l-1)(2x-1)/2 \leq p_B(\cdot) < c$ and increases on $a \leq p_B(\cdot) \leq \underline{t}(2l-1)(2x-1)/2$. Hence, $\Pi_B(\cdot)$ gets its maximum at $p_B^*(x, k) = \underline{t}(2l-1)(2x-1)/2$ and $\Pi_B^{D,ND}(x, k) = \underline{t}(2l-1)^2(2x-1)/[8(l-1)]$. *ii)* $\Pi_B(\cdot)$ increases on the interval $a \leq p_B(\cdot) \leq b$ and decreases on the interval $b \leq p_B(\cdot) \leq c$ if $\log_2 [(l-1)/(l-3/2)] < k < \log_2 [3(l-1)/(2l-4)]$. Hence, $\Pi_B(\cdot)$ gets its maximum at $p_B^*(x, k) = \underline{t} [1 + (l-1)/2^k] (2x-1)$ and $\Pi_B^{D,ND}(x, k) = \underline{t} (1 - 1/2^{k+1}) [1 + (l-1)/2^k] (2x-1)$. *iii)* If $k \geq \log_2 [3(l-1)/(2l-4)]$, then $\Pi_B^{D,ND}(x, k)$ increases on $a \leq p_B(\cdot) \leq \underline{t} [l/2 + (l-1)/2^{k+2}] (2x-1)$ and decreases on $\underline{t} [l/2 + (l-1)/2^{k+2}] (2x-1) \leq p_B(\cdot) \leq c$. Hence, $\Pi_B(\cdot)$ gets its maximum at $p_B^*(x, k) = \underline{t} [l/2 + (l-1)/2^{k+2}] (2x-1)$ and $\Pi_B^{D,ND}(x, k) = \underline{t} [l/2 + (l-1)/2^{k+2}]^2 (2x-1)/(l-1)$.

Note finally that $\log_2 [(l-1)/(l-3/2)] < 1$ holds for any $l > 2$, such that case *i)* is irrelevant for any $k \geq 1$. Also, $\log_2 [3(l-1)/(2l-4)] \leq 1$ if $l \geq 5$. We next compute the profits of firm A . We consider in turn different cases with respect to the optimal price of firm B derived above. Assume first that $l < 5$, then both cases *ii)* and *iii)* apply for $k \geq 1$.

ii) If $1 \leq k < \log_2 [3(l-1)/(2l-4)]$, then $p_B^*(x, k) = \underline{t} [1 + (l-1)/2^k] (2x-1)$. As $p_B^*(x, k) = b$, firm A serves consumers only on $m = 1$. (18) yields $p_{A1}^*(x, k) = \underline{t}(l-1)(2x-1)/2^{k+1}$, and firm A serves consumers with $t < \underline{t} [1 + (l-1)/2^{k+1}]$. The profit of firm A on the turf of firm B is $\Pi_A^{D,ND}(x > 1/2, k) = \underline{t}(l-1)/[2^{2k+4}]$.

iii) If $k \geq \log_2 [3(l-1)/(2l-4)]$, then $p_B^*(x, k) = \underline{t} [l/2 + (l-1)/2^{k+2}] (2x-1)$, where $b \leq p_B^*(x, k) < c$. For any l there is some $\hat{m}$, such that $p_B^*(x, k)/(2x-1) \in [\underline{t}^{\hat{m}}(k), \bar{t}^{\hat{m}}(k)]$. Unfortunately, there is no analytical solution for $\hat{m} \in N$ depending on l . We will assume instead that

$p_B^*(x, k) = \underline{t}^{\widehat{m}}(k)(2x - 1)$, such that $\widehat{m} = 5/4 + 2^k(l - 2) / [2(l - 1)]$. Note that the condition $\widehat{m} \in N$ does not necessarily hold and for any k we are able to derive only the approximate value of $\Pi_A^{D,ND}(x > 1/2, k)$. Given $p_B^*(x, k) = \underline{t}^{\widehat{m}}(k)$, firm B serves all consumers on $m \geq \widehat{m}$, while firm A serves all consumers on $m \leq \widehat{m} - 2$ and some consumers on $m = \widehat{m} - 1$. (18) yields $p_{A, \widehat{m}-1}^*(x, k) = \underline{t}(l - 1)(2x - 1) / 2^{k+1}$, such that on $\widehat{m} - 1$ on firm B 's turf firm A realizes the profit $\Pi_{A, \widehat{m}-1}^{D,ND}(x > 1/2, k) = \underline{t}(l - 1) / 2^{2k+4}$. On $m \leq \widehat{m} - 2$ firm A charges the prices $p_{Am}^*(x, k) = p_B^*(x, k) - \bar{t}^m(k)(2x - 1)$ and profits of firm A on the turf of firm B are

$$\begin{aligned} \Pi_A^{D,ND}(x > 1/2, k) &= \frac{\underline{t}(l - 1)}{2^{2k+4}} + \frac{1}{2^k} \int_{1/2}^1 (2x - 1) dx \sum_1^{\widehat{m}-2} \left(\underline{t}^{\widehat{m}}(k) - \bar{t}^m(k) \right) \\ &= \frac{\underline{t}(l - 1)}{2^{2k+4}} + \frac{\underline{t}}{2^7} \left[\frac{2(l - 2)}{l - 1} + \frac{1}{2^k} \right] \left[2(l - 2) - \frac{3(l - 1)}{2^k} \right]. \end{aligned}$$

Note finally that $\Pi_A^{D,ND}(x > 1/2, k)$ can be expressed as

$$\Pi_A^{D,ND}(x > 1/2, k) = \frac{\underline{t}(l - 1)}{2^{2k+4}} + \frac{\underline{t}(l - 1)(\widehat{m} - 1)(\widehat{m} - 2)}{2^{2k+3}}, \quad (19)$$

such that the second expression in the RHS of (19), the profits of firm A on $m \leq \widehat{m} - 2$, is zero if either $\widehat{m} = 1$ or $\widehat{m} = 2$.

If $l \geq 5$, then only case *iii*) applies for any $k \geq 1$. *Q.E.D.*

Proof of Lemma 8. We will prove the lemma for firm B , the results for firm A follow then directly. Note that $\log_2(l - 1) \geq \log_2[3(l - 1) / (2l - 4)]$ if $l \geq 7/2$ and $\log_2(l - 1) > 1$ if $l > 3$. Finally, $\log_2[3(l - 1) / (2l - 4)] > 1$ if $l < 5$. To compute the total profits of firm B on both turfs we have to consider different cases depending on l .

i) $2 < l \leq 3$. There are two relevant data quality intervals to consider: *a)* if $1 \leq k \leq \log_2[3(l - 1) / (2l - 4)]$, then $\Pi_B^{D,ND}(k) = \Pi_B^2(x < 1/2, k) + \Pi_B^1(x > 1/2, k)$ and *b)* if $k > \log_2[3(l - 1) / (2l - 4)]$, then $\Pi_B^{D,ND}(k) = \Pi_B^2(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$.

ii) $3 < l \leq 7/2$. There are three relevant data quality intervals to consider: *a)* if $1 \leq k < \log_2(l - 1)$, then $\Pi_B^{D,ND}(k) = \Pi_B^1(x < 1/2, k) + \Pi_B^1(x > 1/2, k)$, *b)* if $\log_2(l - 1) \leq k < \log_2[3(l - 1) / (2l - 4)]$, then $\Pi_B^{D,ND}(k) = \Pi_B^2(x < 1/2, k) + \Pi_B^1(x > 1/2, k)$ and *c)* if $k \geq \log_2[3(l - 1) / (2l - 4)]$, then $\Pi_B^{D,ND}(k) = \Pi_B^2(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$.

iii) $7/2 < l < 5$. There are three relevant data quality intervals to consider: *a)* if $1 \leq k < \log_2[3(l - 1) / (2l - 4)]$, then $\Pi_B^{D,ND}(k) = \Pi_B^1(x < 1/2, k) + \Pi_B^1(x > 1/2, k)$, *b)* $\Pi_B^{D,ND}(k) =$

$\Pi_B^1(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$ if $\log_2[3(l-1)/(2l-4)] \leq k < \log_2(l-1)$ and $c)$ if $k \geq \log_2(l-1)$, then $\Pi_B^{D,ND}(k) = \Pi_B^2(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$.

iv) $l \geq 5$. There are two relevant data quality intervals: *a)* if $1 \leq k < \log_2(l-1)$, then $\Pi_B^{D,ND}(k) = \Pi_B^1(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$ and *b)* if $k \geq \log_2(l-1)$, then $\Pi_B^{D,ND}(k) = \Pi_B^2(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$.

All these cases together yield the results stated in Lemma 8. *Q.E.D.*

Proof of Proposition 6. *Part 1: The incentives of firm B to acquire data.* We have to compare $\Pi_B^{D,D}(k)$ and $\Pi_B^{D,ND}(k)$ for different values of k and l . Note first that if $l \leq 3$, then $\log_2(l-1) \leq 1$. Hence, it follows from Lemma 5 that if $l \leq 3$, then $\Pi_i^{D,D}(1) = \Pi_i^2(1) = \underline{t}(l+3)/16$. If $l > 3$, then $\Pi_i^{D,D}(1) = \Pi_i^1(1) = (7l-3)/[72(l-1)]$.

i) If $2 < l \leq 3$, it follows from Lemma 8 that $\Pi_B^{D,ND}(1) = \Pi_B^1(x > 1/2, 1) = 3\underline{t}(l+1)/32$ as $\Pi_B^2(x < 1/2, k) = 0$. We have $\Pi_i^2(1) - \Pi_B^1(x > 1/2, 1) = \underline{t}(3-l)/32 \geq 0$. We know from Lemma 5 that for any $k \geq 1$ it holds $\Pi_i^{D,D}(k) = \Pi_i^2(k)$. As $\Pi_i^2(k)$ increases in k for any $k \geq 1$ and $\Pi_B^{D,ND}(k)$ decreases in k for any $k \geq 1$, we conclude that firm B acquires data of any quality $k \geq 1$.

ii) If $3 < l \leq 7/2$, it follows from Lemma 8 that $\Pi_B^{D,ND}(1) = \Pi_B^1(x < 1/2, 1) + \Pi_B^1(x > 1/2, 1)$, $\Pi_B^1(x < 1/2, 1) = \underline{t}(l-3)^2/[128(l-1)]$ and $\Pi_B^1(x > 1/2, 1) = 3\underline{t}(l+1)/32$. We get

$$\Pi_i^1(1) - \Pi_B^1(x < 1/2, 1) - \Pi_B^1(x > 1/2, 1) = -\frac{\underline{t}(5l+9)(l-3)}{1152(l-1)} < 0.$$

Hence, firm B does not acquire data of quality $k = 1$.

We next show that firm B acquires data of quality $k \geq 2$. Note that $\log_2(l-1) < 2$ if $l < 5$. It follows from Lemma 5 that for any $k \geq 2$ we have $\Pi_i^{D,D}(k) = \Pi_i^2(k)$. We get $\Pi_i^2(\log_2(l-1)) = 1/8$. We know from Lemma 8 that $\Pi_B^{D,ND}(\log_2(l-1)) = \Pi_B^1(x > 1/2, \log_2(l-1))$ as $\Pi_B^2(x < 1/2, k) = 0$, and $\Pi_B^1(x > 1/2, \log_2(l-1)) = \underline{t}(2l-3)/[4(l-1)]$. The comparison yields

$$\Pi_i^2(\log_2(l-1)) - \Pi_B^1(x > 1/2, \log_2(l-1)) = \frac{\underline{t}(l-2)(l-3)}{8(l-1)} > 0.$$

Hence, as $\Pi_i^2(k)$ increases in k on $k \geq \log_2(l-1)$ and $\Pi_B^{D,ND}(k)$ decreases in k for any $k \geq 1$, we conclude that firm B acquires data of any quality $k \geq 2$.

iii) If $7/2 < l < 5$, it follows from Lemma 8 that $\Pi_B^{D,ND}(1) = \Pi_B^1(x < 1/2, 1) + \Pi_B^1(x > 1/2, 1)$. As shown in case *ii)* firm B does not acquire data of quality $k = 1$. We know from Lemma 5

that $\Pi_i^{D,D}(\log_2(l-1)) = \Pi_i^2(\log_2(l-1))$ and $\Pi_i^2(\log_2(l-1)) = 1/8$. We know from Lemma 8 that $\Pi_B^{D,ND}(\log_2(l-1)) = \Pi_B^2(x > 1/2, \log_2(l-1))$ as $\Pi_B^2(x < 1/2, k) = 0$, and $\Pi_B^2(x > 1/2, \log_2(l-1)) = \underline{t}(2l+1)^2 / [64(l-1)]$. The comparison shows

$$\Pi_i^2(\log_2(l-1)) - \Pi_B^2(x > 1/2, \log_2(l-1)) = -\frac{\underline{t}(-4l^2 + 12l + 1)}{64(l-1)} > 0.$$

Since $\log_2(l-1) < 2$ for any $l < 5$, then $\Pi_i^{D,D}(k) = \Pi_i^2(k)$ for any $k \geq 2$, as follows from Lemma 5. As follows from Lemma 8, $\Pi_B^{D,ND}(k) = \Pi_B^2(x > 1/2, k)$ for any $k \geq \log_2(l-1)$ because $\Pi_B^2(x < 1/2, k) = 0$. Since $\Pi_i^2(k)$ increases in k on $k \geq \log_2(l-1)$ and $\Pi_B^2(x > 1/2, k)$ decreases in k for any $k \geq \log_2(l-1)$, we conclude that firm B acquires data of any quality $k \geq 2$.

iv) If $l \geq 5$, then $\log_2(l-1) \geq 2$, hence, according to Lemma 5, $\Pi_i^{D,D}(k) = \Pi_i^1(k)$ if $k = 1$ and $k = 2$ (as $\Pi_i^1(\log_2(l-1)) = \Pi_i^2(\log_2(l-1))$). Also it follows from Lemma 8 that $\Pi_B^{D,ND}(k) = \Pi_B^1(x < 1/2, k) + \Pi_B^2(x > 1/2, k)$ if $k = 1$ and $k = 2$ (note that $\Pi_B^1(x < 1/2, \log_2(l-1)) = \Pi_B^2(x < 1/2, \log_2(l-1))$). We will show next that firm B does not acquire data of quality $k = 1$ and acquires data of quality $k = 2$. We have that $\Pi_i^1(1) = \underline{t}l(7l-3) / [72(l-1)]$ and $\Pi_i^1(2) = -\underline{t}(-59l^2 + 2l + 25) / [576(l-1)]$. We also have $\Pi_B^1(x < 1/2, 1) = \underline{t}(l-3)^2 / [128(l-1)]$ and $\Pi_B^1(x < 1/2, 2) = \underline{t}(l-5)^2 / [512(l-1)]$. We get $\Pi_B^2(x > 1/2, 1) = \underline{t}(5l-1)^2 / [256(l-1)]$ and $\Pi_B^2(x > 1/2, 2) = \underline{t}(9l-1)^2 / [1024(l-1)]$. The comparison shows

$$\Pi_i^1(1) - \Pi_B^1(x < 1/2, 1) - \Pi_B^2(x > 1/2, 1) = -\frac{\underline{t}(19l^2 - 102l + 171)}{2304(l-1)} < 0,$$

hence, firm B does not acquire data of quality $k = 1$. We also have that

$$\Pi_i^1(2) - \Pi_B^1(x < 1/2, 2) - \Pi_B^2(x > 1/2, 2) = \frac{\underline{t}(197l^2 + 310l - 859)}{9216(l-1)} > 0, \quad (20)$$

hence, firm B acquires data of quality $k = 2$. We finally show that firm B acquires data of any quality $k \geq 2$. Note that $\Pi_i^1(k)$ gets its minimum at $k = \log_2[20(l-1)/(9l+5)]$ and $\log_2[20(l-1)/(9l+5)] < 2$ for any l ($\log_2[20(l-1)/(9l+5)] \geq 0$ for any $l \geq 25/11$). Also, $\Pi_i^2(k)$ increases in k on $k \geq \log_2(l-1)$. Hence, $\Pi_i^{D,D}(k)$ increases in k on $k \geq 2$. As $\Pi_B^{D,ND}(k)$ decreases in k for any $k \geq 1$, it follows from (20) that firm B acquires data of any quality $k \geq 2$.

Combining the results of our analysis in cases *i*)-*iv*) we conclude that if $l \leq 3$, then firm B

acquires data of any quality $k \geq 1$. If $l > 3$, then firm B does not acquire data of quality $k = 1$ and acquires data of quality $k \geq 2$.

Part 2: The incentives of firm A to acquire data. We have to compare $\Pi_i^{D,D}(0)$ with $\Pi_A^{D,ND}(k)$ for different k and l . Note that for any $l > 2$ it holds $\log_2(l-1) > 0$, hence, according to Lemma 5, $\Pi_i^{D,D}(0) = \Pi_i^1(0) = \underline{t}(5l^2 - 8l + 5) / [36(l-1)]$.

i) If $2 < l \leq 3$, then according to Lemma 8, for $1 \leq k \leq \log_2[3(l-1)/(2(l-2))]$ we have $\Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, k) + \Pi_A^1(x > 1/2, k)$, which increases in k for any $k \geq 0$. If $k > \log_2[3(l-1)/(2(l-2))]$, then $\Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$, which increases in k if $k > \log_2[5(l-1)/(2(5l-6))]$. Note that for any $l > 1$ it holds $\log_2[5(l-1)/(2(5l-6))] < \log_2[3(l-1)/(2(l-2))]$, hence, $\Pi_A^2(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$ increases in k for any $k > \log_2[3(l-1)/(2(l-2))]$. We conclude that $\Pi_A^{D,ND}(k)$ increases in k for any $k \geq 1$ and, hence, $\min_k \Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, 1) + \Pi_A^2(x > 1/2, 1)$. The comparison shows

$$\Pi_A^2(x < 1/2, 1) + \Pi_A^1(x > 1/2, 1) - \Pi_i^1(0) = -\frac{\underline{t}(35l^2 - 182l + 179)}{576(l-1)} > 0,$$

such that firm A acquires data of quality $k = 1$. We conclude that firm A acquires data of any quality $k \geq 1$.

ii) If $3 < l \leq 7/2$, then according to Lemma 8, for $1 \leq k < \log_2(l-1)$ we have $\Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, k) + \Pi_A^1(x > 1/2, k)$. The latter function decreases in k if $k < \log_2[5(l-1)]$. As $\log_2(l-1) < \log_2[5(l-1)]$ for any $l > 1$, then $\Pi_A^{D,ND}(k)$ decreases in k for any $1 \leq k < \log_2(l-1)$. If $\log_2(l-1) \leq k < \log_2[3(l-1)/(2(l-2))]$, then according to Lemma 8, $\Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, k) + \Pi_A^1(x > 1/2, k)$, which increases for any $k \geq 0$. Finally, if $k \geq \log_2[3(l-1)/(2(l-2))]$, then $\Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$, which increases in k if $k > \log_2[5(l-1)/(2(5l-6))]$. As for any $l > 1$ it holds that $\log_2[5(l-1)/(2(5l-6))] < \log_2[3(l-1)/(2(l-2))]$, then $\Pi_A^{D,ND}(k)$ increases in k for any $k \geq \log_2[3(l-1)/(2(l-2))]$. We conclude that $\min_k \Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, \log_2(l-1)) + \Pi_A^1(x > 1/2, \log_2(l-1))$. The comparison shows

$$\Pi_A^2(x < 1/2, \log_2(l-1)) + \Pi_A^1(x > 1/2, \log_2(l-1)) - \Pi_i^1(0) = -\frac{\underline{t}(2l^2 - 14l + 11)}{144(l-1)} > 0,$$

such that firm A acquires data of quality $k = \log_2(l-1)$. We conclude that firm A acquires data of any quality $k \geq 1$.

iii) If $7/2 < l < 5$, then according to Lemma 8, for $1 \leq k < \log_2 [3(l-1)/(2(l-2))]$ we have $\Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, k) + \Pi_A^1(x > 1/2, k)$, which decreases in k if $k < \log_2 [5(l-1)]$. Note that $\log_2 [3(l-1)/(2(l-2))] < \log_2 [5(l-1)]$ for any $l > 23/10$. Hence, $\Pi_A^{D,ND}(k)$ decreases in k for $1 \leq k < \log_2 [3(l-1)/(2(l-2))]$. According to Lemma 8, if $\log_2 [3(l-1)/(2(l-2))] \leq k < \log_2 (l-1)$, then $\Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$, which increases in k if $k > \log_2 (7/2)$. Note that for any $l > 11/4$ it holds that $\log_2 [3(l-1)/(2(l-2))] < \log_2 (7/2)$. Also, $\log_2 (l-1) \leq \log_2 (7/2)$ if $l \leq 9/2$. It follows that if $7/2 < l \leq 9/2$, then $\Pi_A^1(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$ decreases for any $\log_2 [3(l-1)/(2(l-2))] \leq k < \log_2 (l-1)$. If $9/2 < l < 5$, then $\Pi_A^{D,ND}(k)$ decreases on $\log_2 [3(l-1)/(2(l-2))] \leq k \leq \log_2 (7/2)$ and increases on $\log_2 (7/2) < k < \log_2 (l-1)$. Consider finally $k \geq \log_2 (l-1)$, where according to Lemma 8 $\Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$, which increases in k if $k > \log_2 [5(l-1)/(2(5l-6))]$. As for any $l > 17/10$ it holds that $\log_2 [5(l-1)/(2(5l-6))] < \log_2 (l-1)$, then $\Pi_A^{D,ND}(k)$ increases in k for any $k \geq \log_2 (l-1)$. We conclude that if $7/2 < l \leq 9/2$, then $\min_k \Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, \log_2 (l-1)) + \Pi_A^2(x > 1/2, \log_2 (l-1))$ and if $9/2 < l < 5$, then $\min_k \Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, \log_2 (7/2)) + \Pi_A^2(x > 1/2, \log_2 (7/2))$. The comparison shows that if $7/2 < l \leq 9/2$, then

$$\Pi_A^1(x < 1/2, \log_2 (l-1)) + \Pi_A^2(x > 1/2, \log_2 (l-1)) - \Pi_i^1(0) = \frac{t(20l^2 - 68l + 101)}{1152(l-1)} > 0,$$

such that firm A acquires data of quality $k = \log_2 (l-1)$. And if $9/2 < l < 5$, then

$$\Pi_A^1(x < 1/2, \log_2 (7/2)) + \Pi_A^2(x > 1/2, \log_2 (7/2)) - \Pi_i^1(0) = -\frac{t(-52l^2 + 76l + 11)}{4032(l-1)} > 0,$$

such that firm A acquires data of quality $k = \log_2 (7/2)$. We conclude that firm A acquires data of any quality $k \geq 1$.

iv) If $l \geq 5$, then according to Lemma 8 for $1 \leq k < \log_2 (l-1)$ we have $\Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$, which increases in k if $k > \log_2 (7/2)$. Note that for any $l \geq 5$ it holds that $1 < \log_2 (7/2) < \log_2 (l-1)$, hence, $\Pi_A^{D,ND}(k)$ gets its minimum on $1 \leq k < \log_2 (l-1)$ at $k = \log_2 (7/2)$. If $k \geq \log_2 (l-1)$, then $\Pi_A^{D,ND}(k) = \Pi_A^2(x < 1/2, k) + \Pi_A^2(x > 1/2, k)$, which increases in k if $k > \log_2 [5(l-1)/(2(5l-6))]$. As for any $l > 2$ it holds that $\log_2 [5(l-1)/(2(5l-6))] < \log_2 (l-1)$, $\Pi_A^{D,ND}(k)$ increases in k if $k \geq \log_2 (l-1)$. We con-

clude that $\min_k \Pi_A^{D,ND}(k) = \Pi_A^1(x < 1/2, \log_2(7/2)) + \Pi_A^2(x > 1/2, \log_2(7/2))$. We get

$$\Pi_A^1(x < 1/2, \log_2(7/2)) + \Pi_A^2(x > 1/2, \log_2(7/2)) - \Pi_i^1(0) = -\frac{\underline{t}(-52l^2 + 76l + 11)}{4032(l-1)} > 0,$$

hence, firm A acquires data of quality $k = \log_2(7/2)$. It follows that firm A acquires data of any quality $k \geq 1$.

Combining the results of our analysis in cases i)- iv) we conclude that firm A acquires data of any quality $k \geq 1$ for any $l > 2$. Combining the incentives of firms A and B to acquire customer data, we get the following equilibria. If $l \leq 3$, a unique Nash equilibrium (in dominant strategies) exists, where each firm acquires customer data of any quality $k \geq 1$. If $l > 3$, the equilibrium depends on the quality of customer data: i) if $k = 1$, then two Nash equilibria exist where one of the firms acquires data, ii) if $k \geq 2$, then the unique Nash equilibrium (in dominant strategies) exists, where both firms acquire data. *Q.E.D.*

Proof of Proposition 7. If $l \leq 3$, then both firms acquire data of any quality $k \geq 1$. For any $l \leq 3$ it holds that $\log_2(l-1) \leq 1$, hence, according to Lemma 5, if firms acquire data $k \geq 1$, they get the profit $\Pi_i^{D,D}(k) = \Pi_i^2(k)$. We have $\Pi_i^2(1) = \underline{t}(l+3)/16$. For any $l \leq 3$ it holds $\log_2(l-1) > 0$. Hence, according to Lemma 5, if firms do not acquire data, they realize $\Pi_i^{D,D}(0) = \Pi_i^1(0) = \underline{t}(5l^2 - 8l + 5)/[36(l-1)]$. We get $\Pi_i^2(1) - \Pi_i^1(0) = -\underline{t}(11l^2 - 50l + 47)/[144(l-1)] > 0$, hence, firms are better off following the acquisition of data of quality $k = 1$. As $\Pi_i^2(k)$ increases in k for any $k \geq 1$, we conclude that firms are better off following the acquisition of data of any quality $k \geq 1$.

We next consider the case $l > 3$ and $k = 1$, where only firm A acquires data. It is straightforward that the profit of firm A increases. We next analyze how the profit of firm B changes. If $l < 5$, then according to Lemma 8, $\Pi_B^{D,ND}(1) = \Pi_B^1(x < 1/2, 1) + \Pi_B^1(x > 1/2, 1) = -\underline{t}(-13l^2 + 6l + 3)/[128(l-1)]$. As $\log_2(l-1) > 0$ for any $l > 2$, then $\Pi_i^{D,D}(0) = \Pi_i^1(0) = \underline{t}(5l^2 - 8l + 5)/[36(l-1)]$. The comparison shows

$$\Pi_B^1(x < 1/2, 1) + \Pi_B^1(x > 1/2, 1) - \Pi_i^1(0) = -\frac{\underline{t}(43l^2 - 202l + 187)}{1152(l-1)}. \quad (21)$$

Note that the RHS of (21) is positive if $3 < l \leq (12\sqrt{15} + 101)/43$, in which case firm B (weakly) gains from data acquisition, and negative if $(12\sqrt{15} + 101)/43 < l < 5$, in which case firm B loses from data acquisition. If $l \geq 5$, then according to Lemma 8, $\Pi_B^{D,ND}(1) = \Pi_B^1(x <$

$1/2, 1) + \Pi_B^2(x > 1/2, 1) = \underline{t} (27l^2 - 22l + 19) / [256(l - 1)]$. As above, $\Pi_i^{D,D}(0) = \Pi_i^1(0) = \underline{t} (5l^2 - 8l + 5) / [36(l - 1)]$. The comparison shows

$$\Pi_B^1(x < 1/2, 1) + \Pi_B^2(x > 1/2, 1) - \Pi_i^1(0) = -\frac{\underline{t} (77l^2 - 314l + 149)}{2304(l - 1)} < 0.$$

We conclude that firm B is worse off with data acquisition if $l \geq 5$.

We finally consider the case $l > 3$ and $k \geq 2$, where both firms acquire data. As above, $\Pi_i^{D,D}(0) = \Pi_i^1(0) = \underline{t} (5l^2 - 8l + 5) / [36(l - 1)]$. Note that if $l \leq 5$, then $\log_2(l - 1) \leq 2$, hence, according to Lemma 5, for any $k \geq 2$ we have $\Pi_i^{D,D}(k) = \Pi_i^2(k)$. We have $\Pi_i^2(2) = \underline{t} (3l + 5) / 32$. We get $\Pi_i^2(2) - \Pi_i^1(0) = -\underline{t}(l - 5)(13l - 17) / [288(l - 1)] \geq 0$ if $l \leq 5$. As $\Pi_i^2(k)$ increases in k for $k \geq 2$, we conclude that if $l \leq 5$, then firms are (weakly) better off following the acquisition of data of any quality $k \geq 2$.

In the following we will assume that $l > 5$. According to Lemma 5, if $2 \leq k < \log_2(l - 1)$, then $\Pi_i^{D,D}(k) = \Pi_i^1(k)$ and if $k \geq \log_2(l - 1)$, then $\Pi_i^{D,D}(k) = \Pi_i^2(k)$, which increases in k for any $k \geq \log_2(l - 1)$. Note that $\Pi_i^1(k)$ decreases in k if $k < \log_2[20(l - 1) / (9l + 5)]$. For any l it holds that $\log_2[20(l - 1) / (9l + 5)] < 2$. We conclude that $\Pi_i^{D,D}(k)$ increases in k for any $k \geq 2$ if $l > 5$. As above, $\Pi_i^{D,D}(0) = \Pi_i^1(0) = \underline{t} (5l^2 - 8l + 5) / [36(l - 1)]$. To conclude on the profit effect of data acquisition, we have to consider the differences $\Pi_i^1(0) - \Pi_i^1(k)$ and $\Pi_i^1(0) - \Pi_i^2(k)$. We get $\Pi_i^1(0) - \Pi_i^1(k) = -(2^k - 1) [(15 - l) 2^k - 10(l - 1)] / (72 \times 2^{2k})$. Note that $(15 - l) 2^k - 10(l - 1) < 0$ if $l < 15$ and $k < \log_2[10(l - 1) / (15 - l)]$ or if $l \geq 15$. Note also that for any $l > 5$ it holds $\log_2[10(l - 1) / (15 - l)] > \log_2(l - 1)$, which implies that for any $2 \leq k < \log_2(l - 1)$ we have that $k < \log_2[10(l - 1) / (15 - l)]$. We conclude that for any $l > 5$ and any $2 \leq k < \log_2(l - 1)$ we have $(15 - l) 2^k - 10(l - 1) < 0$, such that $\Pi_i^1(0) > \Pi_i^1(k)$ and firms are worse off when they acquire data of quality $2 \leq k < \log_2(l - 1)$. We get

$$\Pi_i^1(0) - \Pi_i^2(k) = \frac{\underline{t} [2^k(l^2 - 16l + 19) + 9l^2 - 18l + 9]}{72 \times 2^k(l - 1)}.$$

Note that $9l^2 - 18l + 9 > 0$ for any $l > 5$ and $l^2 - 16l + 19 \geq 0$ if $l \geq 8 + 3\sqrt{5}$. We conclude that if $l \geq 8 + 3\sqrt{5}$, then $\Pi_i^1(0) - \Pi_i^2(k) > 0$ for any k , such that firms are worse off following the acquisition of data of any quality $k \geq \log_2(l - 1)$. Consider now $l < 8 + 3\sqrt{5}$, in which case

$\Pi_i^1(0) > \Pi_i^2(k)$ if $k < \log_2 \left[- (9l^2 - 18l + 9) / (l^2 - 16l + 19) \right]$. The comparison shows that

$$-\frac{9l^2 - 18l + 9}{l^2 - 16l + 19} - (l - 1) = -\frac{(l - 1)(l - 5)(l - 2)}{[l - (3\sqrt{5} + 8)][l - (8 - 3\sqrt{5})]} > 0 \text{ if } 5 < l < 8 + 3\sqrt{5}.$$

We conclude that given $5 < l < 8 + 3\sqrt{5}$ firm are worse off following data acquisition if $\log_2(l - 1) \leq k < \log_2 \left[- (9l^2 - 18l + 9) / (l^2 - 16l + 19) \right]$ and are (weakly) better off if $k \geq \log_2 \left[- (9l^2 - 18l + 9) / (l^2 - 16l + 19) \right]$. *Q.E.D.*

Proof of Proposition 8. We start with social welfare. Assume that $k < \log_2(l - 1)$. From Lemma 5 we know that every firm serves consumers with $t \geq \underline{t} \left[2 + (l - 1) / 2^k \right] / 3$ on its turf, such that social welfare can be computed as

$$\begin{aligned} SW(k) &= v - 2 \int_{\frac{2\underline{t}}{3} + \frac{1-\underline{t}}{3 \times 2^k}}^1 f_t \int_0^{1/2} (tx) dx dt - 2 \int_{\underline{t}}^{\frac{2\underline{t}}{3} + \frac{1-\underline{t}}{3 \times 2^k}} f_t \int_0^{1/2} [t(1 - x)] dx dt \\ &= SW^1(k) := v - \frac{\underline{t}(l^2 - 3)}{8(l - 1)} - \frac{\underline{t}}{4(l - 1)} \left(\frac{2}{3} + \frac{l - 1}{3 \times 2^k} \right)^2. \end{aligned}$$

We know from Lemma 5 that if $k \geq \log_2(l - 1)$, then every firm serves all consumers on its turf, such that $SW = v - 2 \int_{\underline{t}}^1 f_t \int_0^{1/2} (tx) dx dt = SW^2 := v - \underline{t}(1 + l) / 8$. If firms do not acquire data, then social welfare is given by $SW^1(0)$. As $SW^1(\log_2(l - 1)) = SW^2$ and $SW^1(k)$ increases in k , we conclude that social welfare always increases when both firms acquire data.

We next analyze how consumers surplus changes when both firms acquire data in equilibrium. To compute consumer surplus we will consider the difference between social welfare and firms' profits stated in Lemma 5. If $k < \log_2(l - 1)$, then

$$\begin{aligned} CS(k) &= SW^1(k) - 2\Pi_i^1(k) \\ &= CS^1(k) = v - \frac{\underline{t} \left[2^{2k} (27l^2 - 29) + 2^k (16l - 18l^2 + 2) - 44l + 22l^2 + 22 \right]}{72 \times 2^{2k} (l - 1)}. \end{aligned}$$

We show that $CS^1(k)$ increases in k . Consider $f_1(k, l) := 2^{2k} (27l^2 - 29) + 2^k (16l - 18l^2 + 2) - 44l + 22l^2 + 22$. We get $\partial f_1(k, l) / \partial k = 2^{k+1} (\ln 2) \left[2^k (27l^2 - 29) - (l - 1)(9l + 1) \right]$. We next consider $f_2(k, l) := 2^k (27l^2 - 29) - (l - 1)(9l + 1)$. Note that $27l^2 - 29 > 0$ for any $l > 2$, such that $f_2(k, l) > 0$ if $k > \log_2 \left[(l - 1)(9l + 1) / (27l^2 - 29) \right]$. For any $l > 2$ it holds $(l - 1)(9l + 1) / (27l^2 - 29) < 1$, such that $\log_2 \left[(l - 1)(9l + 1) / (27l^2 - 29) \right] < 0$. We conclude that for any $l > 2$ and any $0 \leq k < \log_2(l - 1)$ it holds $f_2(k, l) > 0$, such that $\partial f_1(k, l) / \partial k > 0$

and $CS^1(k)$ decreases in k . If $k \geq \log_2(l-1)$, then

$$CS(k) = SW^2 - 2\Pi_i^2(k) = CS^2(k) := v - 3\underline{t}(1+l)/8 + \underline{t}(l-1)/2^{2+k}.$$

It is straightforward that $CS^2(k)$ decreases in k . As $CS^1(\log_2(l-1)) = CS^2(\log_2(l-1))$, we conclude that $CS(k)$ increases on $k < \log_2(l-1)$ and decreases on $k \geq \log_2(l-1)$. Note that $CS^1(0) = v + \underline{t}(-31l^2 + 28l + 5)/[72(l-1)]$. We get

$$CS^1(0) - CS^2(k) = \underline{t}(-2l^2 + 14l - 11)/[36(l-1)] - \underline{t}(l-1)/2^{2+k}. \quad (22)$$

First note that $-2l^2 + 14l - 11 \leq 0$ if $l \geq (7 + 3\sqrt{3})/2$, in which case it follows from (22) that $CS^1(0) < CS^2(k)$ for any $k \geq \log_2(l-1)$. We conclude that if $l \geq (7 + 3\sqrt{3})/2$, then consumers are better off when firms acquire data of any quality. Assume next that $l < (7 + 3\sqrt{3})/2$, in which case $-2l^2 + 14l - 11 > 0$.

The RHS of (22) is positive if $k > \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$. We get

$$9(l-1)^2 / (-2l^2 + 14l - 11) - (l-1) = (l-2)(l-1)(2l-1) / (-2l^2 + 14l - 11) > 0,$$

which implies that $\log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right] > \log_2(l-1)$. Hence, $CS^1(0) > CS^2(k)$ if $k > \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$. Otherwise, $CS^1(0) \leq CS^2(k)$ if $\log_2(l-1) \leq k \leq \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$. We conclude that if $l < (7 + 3\sqrt{3})/2$, the effect of data acquisition on consumer surplus depends on its quality. If $k \leq \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$, then consumers are (weakly) better off following data acquisition. Consumers are worse off following data acquisition if $k > \log_2 \left[9(l-1)^2 / (-2l^2 + 14l - 11) \right]$. *Q.E.D.*

Proof of Lemma 9. Maximization of firms' profits on some $x < 1/2$ and segment m yields the best-response functions

$$p_{Am}(x, k, p_B) = \begin{cases} \frac{\bar{t}^m(k)(1-2x) + p_B}{2} & \text{if } p_B \leq [\bar{t}^m(k) - 2\underline{t}^m(k)](1-2x) \\ \underline{t}^m(k)(1-2x) + p_B & \text{if } p_B > [\bar{t}^m(k) - 2\underline{t}^m(k)](1-2x) \end{cases} \quad (23)$$

and

$$p_{Bm}(x, k, p_A) = \begin{cases} p_A - \bar{t}^m(k)(1-2x) & \text{if } \frac{p_A}{1-2x} \geq 2\bar{t}^m(k) - \underline{t}^m(k) \\ \frac{p_A - \underline{t}^m(k)(1-2x)}{2} & \text{if } \underline{t}^m(k) < \frac{p_A}{1-2x} < 2\bar{t}^m(k) - \underline{t}^m(k) \\ 0 & \text{if } \frac{p_A}{1-2x} \leq \underline{t}^m(k). \end{cases} \quad (24)$$

We will show first that for any p_B firm A charges a higher price on $m^1 + 1$, which is a strong market of firm A , while segment m^1 is a weak market. Note that $\bar{t}^{m^1}(k) - 2\underline{t}^{m^1}(k) = -\underline{t} + (\bar{t} - \underline{t})(2 - m^1)/2^k$ and $\bar{t}^{m^1+1}(k) - 2\underline{t}^{m^1+1}(k) = -\underline{t} + (\bar{t} - \underline{t})(1 - m^1)/2^k$, for any m^1 it holds that $\bar{t}^{m^1+1}(k) - 2\underline{t}^{m^1+1}(k) < \bar{t}^{m^1}(k) - 2\underline{t}^{m^1}(k)$. We will first assume that $\bar{t}^{m^1+1}(k) - 2\underline{t}^{m^1+1}(k) > 0$ and consider different possibilities for the price of firm B . *i)* Assume that $p_B/(1-2x) \leq \bar{t}^{m^1+1}(k) - 2\underline{t}^{m^1+1}(k)$. Then according to (23), $p_{A,m^1+1}(x, k, p_B) = [\bar{t}^{m^1+1}(k)(1-2x) + p_B]/2$ and $p_{A,m^1}(x, k, p_B) = [\bar{t}^{m^1}(k)(1-2x) + p_B]/2$, and $p_{A,m^1}(x, k, p_B) < p_{A,m^1+1}(x, k, p_B)$. *ii)* Assume that $\bar{t}^{m^1+1}(k) - 2\underline{t}^{m^1+1}(k) < p_B/(1-2x) \leq \bar{t}^{m^1}(k) - 2\underline{t}^{m^1}(k)$. Then $p_{A,m^1+1}(x, k, p_B) = \underline{t}^{m^1+1}(k)(1-2x) + p_B$ and $p_{A,m^1}(x, k, p_B) = [\bar{t}^{m^1}(k)(1-2x) + p_B]/2$. We get

$$[\bar{t}^{m^1}(k)(1-2x) + p_B]/2 - \underline{t}^{m^1+1}(k)(1-2x) - p_B = -(1-2x) \left[\underline{t} + (\bar{t} - \underline{t})m^1/2^k \right] - p_B/2 < 0,$$

which implies $p_{A,m^1}(x, k, p_B) < p_{A,m^1+1}(x, k, p_B)$. *iii)* Assume finally that $p_B/(1-2x) > \bar{t}^{m^1}(k) - 2\underline{t}^{m^1}(k)$. Then $p_{A,m^1+1}(x, k, p_B) = \underline{t}^{m^1+1}(k)(1-2x) + p_B$ and $p_{A,m^1}(x, k, p_B) = \underline{t}^{m^1}(k)(1-2x) + p_B$. It is straightforward that $p_{A,m^1}(x, k, p_B) < p_{A,m^1+1}(x, k, p_B)$. We can conclude that for any p_B it holds that $p_{A,m^1}(x, k, p_B) < p_{A,m^1+1}(x, k, p_B)$, such that segment $m^1 + 1$ is the strong market of firm A . If either $\bar{t}^{m^1+1}(k) - 2\underline{t}^{m^1+1}(k) < 0$ or $\bar{t}^{m^1}(k) - 2\underline{t}^{m^1}(k) < 0$ holds, our conclusions remain unchanged.

We now turn to firm B . We note that for any m it holds that $2\bar{t}^m(k) - \underline{t}^m(k) = \bar{t}^{m+1}(k)$. We conclude that $\underline{t}^{m^1}(k) < \underline{t}^{m^1+1}(k) < 2\bar{t}^{m^1}(k) - \underline{t}^{m^1}(k) < 2\bar{t}^{m^1+1}(k) - \underline{t}^{m^1+1}(k)$. We will consider in turn four cases depending on the price of firm A . *i)* Assume that $p_A/(1-2x) \leq \underline{t}^{m^1}(k)$. Then $p_{Bm^1}(x, k, p_A) = p_{B,m^1+1}(x, k, p_A) = 0$. *ii)* Let $\underline{t}^{m^1}(k) < p_A/(1-2x) \leq \underline{t}^{m^1+1}(k)$. Then $p_{B,m^1+1}(x, k, p_A) = 0$ and $p_{Bm^1}(x, k, p_A) = [p_A - \underline{t}^{m^1}(k)(1-2x)]/2$, and $p_{Bm^1}(x, k, p_A) > p_{B,m^1+1}(x, k, p_A)$. *iii)* Assume that $\underline{t}^{m^1+1}(k) < p_A/(1-2x) < 2\bar{t}^{m^1}(k) - \underline{t}^{m^1}(k)$. Then $p_{B,m^1+1}(x, k, p_A) = [p_A - \underline{t}^{m^1+1}(k)(1-2x)]/2$ and $p_{Bm^1}(x, k, p_A) = p_A/2 - \underline{t}^{m^1}(k)(1-2x)/2$. As $\underline{t}^{m^1+1}(k) > \underline{t}^{m^1}(k)$, we have that $p_{Bm^1}(x, k, p_A) > p_{B,m^1+1}(x, k, p_A)$. *iv)* Let $2\bar{t}^{m^1}(k) - \underline{t}^{m^1}(k) \leq p_A/(1-2x) < 2\bar{t}^{m^1+1}(k) - \underline{t}^{m^1+1}(k)$. Then $p_{Bm^1}(x, k, p_A) =$

$p_A - \bar{t}^{m^1}(k)(1-2x)$ and $p_{B,m^1+1}(x,k,p_A) = [p_A - \underline{t}^{m^1+1}(k)(1-2x)]/2$. We get

$$p_A - \bar{t}^{m^1}(k)(1-2x) - [p_A - \underline{t}^{m^1+1}(k)(1-2x)]/2 = [p_A - (1-2x)\bar{t}^{m^1}(k)]/2 > 0,$$

because $p_A/(1-2x) \geq 2\bar{t}^{m^1}(k) - \underline{t}^{m^1}(k) = \bar{t}^{m^1+1}(k) > \bar{t}^{m^1}(k)$. Hence, $p_{B,m^1}(x,k,p_A) > p_{B,m^1+1}(x,k,p_A)$. *v*) Assume that $p_A/(1-2x) \geq 2\bar{t}^{m^1+1}(k) - \underline{t}^{m^1+1}(k)$. Then $p_{B,m^1}(x,k,p_A) = p_A - \bar{t}^{m^1}(k)(1-2x)$ and $p_{B,m^1+1}(x,k,p_A) = p_A - \bar{t}^{m^1+1}(k)(1-2x)$, such that $p_{B,m^1}(x,k,p_A) > p_{B,m^1+1}(x,k,p_A)$ as $\bar{t}^{m^1}(k) < \bar{t}^{m^1+1}(k)$. We conclude that for any p_A we have $p_{B,m^1}(x,k,p_A) \geq p_{B,m^1+1}(x,k,p_A)$, hence, segment m^1 is the strong market of firm B .

We showed that for any p_B we have $p_{A,m^1}(x,k,p_B) < p_{A,m^1+1}(x,k,p_B)$ and for any p_A we have $p_{B,m^1}(x,k,p_A) \geq p_{B,m^1+1}(x,k,p_A)$. We conclude that firms' best-response functions are characterized by best-response asymmetry. *Q.E.D.*

Proof of Lemma 10. Consider the turf of firm A and segment m with $t \in [\underline{t}^m(k); \bar{t}^m(k)]$. We first show that in equilibrium firm A loses consumers on m . Assume, in contrast, that firm B does not serve any consumers there. Given p_{Bm}^* , firm A sets the highest price allowing to monopolize m , at which the most flexible consumer $(x, t) = (1/2, \underline{t}^m(k))$ is indifferent: $p_{Am}^* = p_{Bm}^*$. It must hold then that $p_{Bm}^* = 0$. (Otherwise, firm B would decrease its price to gain consumers on m .) But then $p_{Am}^* = 0$ must also hold and firm A would deviate to $p_{Am} > 0$.

The transport cost of the indifferent consumer on any $x < 1/2$ is $\tilde{t}_m(x; p_{Am}, p_{Bm}) := (p_{Am} - p_{Bm})/(1-2x)$. Two cases are possible in equilibrium: $\tilde{t}_m(0; p_{Am}^*, p_{Bm}^*) \leq \underline{t}^m(k)$ and $\tilde{t}_m(0; p_{Am}^*, p_{Bm}^*) > \underline{t}^m(k)$. Assume that in equilibrium $\tilde{t}_m(0; p_{Am}^*, p_{Bm}^*) \leq \underline{t}^m(k)$ holds. Then the profit of firm A on m on its turf is

$$\begin{aligned} & \Pi_{Am}(x < 1/2; p_{Am}, p_{Bm}) \\ &= f_t p_{Am} \left[\left(\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\underline{t}^m} \right) (\bar{t}^m - \underline{t}^m) + \int_{\frac{\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\underline{t}^m}}^{\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\bar{t}^m}}} \int_{\frac{p_{Am} - p_{Bm}}{1-2x}}^{\bar{t}^m} dt dx \right] \\ &= f_t p_{Am} \left[\left(\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\underline{t}^m} \right) (\bar{t}^m - \underline{t}^m) + \frac{(p_{Am} - p_{Bm})(\bar{t}^m - 1 - \ln \bar{t}^m)}{2} \right]. \end{aligned}$$

Taking the derivative of $\Pi_{Am}(\cdot)$ with respect to p_{Am} we get the FOC: $p_{Am}(p_{Bm}) = p_{Bm}/2 +$

$(\bar{t}^m - \underline{t}^m) / (2 \ln l^m)$. The profit of firm B on the turf of firm A on m is

$$\begin{aligned}\Pi_{Bm}(x < 1/2; p_{Am}, p_{Bm}) &= f_t p_{Bm} \left[\frac{\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\bar{t}^m} \frac{p_{Am} - p_{Bm}}{1-2x}}{\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\underline{t}^m}} \int_{\underline{t}^m}^{\bar{t}^m} dt dx + \frac{(p_{Am} - p_{Bm})(\bar{t}^m - \underline{t}^m)}{2\bar{t}^m} \right] \\ &= \frac{f_t p_{Bm} (p_{Am} - p_{Bm}) \ln l^m}{2}.\end{aligned}$$

Taking the derivative of $\Pi_{Bm}(\cdot)$ with respect to p_B we get the FOC: $p_{Bm}(p_{Am}) = p_{Am}/2$. Combining the FOCs of the firms we get the prices $p_{Am}^*(x < 1/2, k) = 2[\bar{t}^m(k) - \underline{t}^m(k)] / [3 \ln l^m(k)]$ and $p_{Bm}^*(x < 1/2, k) = [\bar{t}^m(k) - \underline{t}^m(k)] / [3 \ln l^m(k)]$. Firm A serves consumers with $t(x) \geq [\bar{t}^m(k) - \underline{t}^m(k)] / [3 \ln l^m(k)(1-2x)]$. Firms' equilibrium profits are

$$\Pi_{Am}(x < 1/2, k) = \frac{4f_t [\bar{t}^m(k) - \underline{t}^m(k)]^2}{18 \ln l^m(k)} \text{ and } \Pi_{Bm}(x < 1/2, k) = \frac{f_t [\bar{t}^m(k) - \underline{t}^m(k)]^2}{18 \ln l^m(k)}.$$

Condition $\tilde{t}_m(0; p_{Am}^*, p_{Bm}^*) \leq \underline{t}^m(k)$ implies that $3 \ln l^m(k) - l^m(k) + 1 \geq 0$, which yields $l^m(k) \lesssim 6.7$. Hence, if consumers are relatively homogeneous in flexibility, firm A follows a monopolization strategy in equilibrium.

Assume next that $l^m(k) > 6.7$, in which case firm A follows a market-sharing strategy on m on its turf and its profit there is

$$\begin{aligned}\Pi_{Am}(x < 1/2; p_{Am}, p_{Bm}) / f_t &= \\ p_{Am} \int_0^{\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\bar{t}^m}} \int_{\frac{p_{Am} - p_{Bm}}{1-2x}}^{\bar{t}^m} dt dx &= p_{Am} \left[\frac{\bar{t}^m}{2} - \frac{p_{Am} - p_{Bm}}{2} + \frac{(p_{Am} - p_{Bm})}{2} \ln \left(\frac{p_{Am} - p_{Bm}}{\bar{t}^m} \right) \right].\end{aligned}$$

Taking the derivative of $\Pi_{Am}(\cdot)$ with respect to p_{Am} we get the FOC:

$$\bar{t}^m - p_{Am}^* + p_{Bm}^* + (2p_{Am}^* - p_{Bm}^*) \ln \left(\frac{p_{Am}^* - p_{Bm}^*}{\bar{t}^m} \right) = 0.$$

The profit of firm B is

$$\begin{aligned}\Pi_{Bm}(x < 1/2; p_{Am}, p_{Bm}) &= f_t p_{Bm} \left[\frac{\bar{t}^m - \underline{t}^m}{2} - \int_0^{\frac{1}{2} - \frac{p_{Am} - p_{Bm}}{2\bar{t}^m}} \int_{\frac{p_{Am} - p_{Bm}}{1-2x}}^{\bar{t}^m} dt dx \right] \\ &= f_t p_{Bm} \left[\frac{p_{Am} - p_{Bm}}{2} - \frac{\underline{t}^m}{2} - \frac{(p_{Am} - p_{Bm})}{2} \ln \left(\frac{p_{Am} - p_{Bm}}{\bar{t}^m} \right) \right].\end{aligned}$$

Taking the derivative of $\Pi_{Bm}(\cdot)$ with respect to p_{Bm} we get the FOC

$$p_{Am}^* - p_{Bm}^* - \underline{t}^m - (p_{Am}^* - 2p_{Bm}^*) \ln \left(\frac{p_{Am}^* - p_{Bm}^*}{\underline{t}^m} \right) = 0.$$

Q.E.D.

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